Residential signal in frozen embeddings

Compact · partly shared · geographically brittle

Standardized experimental protocol

10 metros

Downtown core boxes

1.75M cells

Balanced 1:1 RES vs NRES

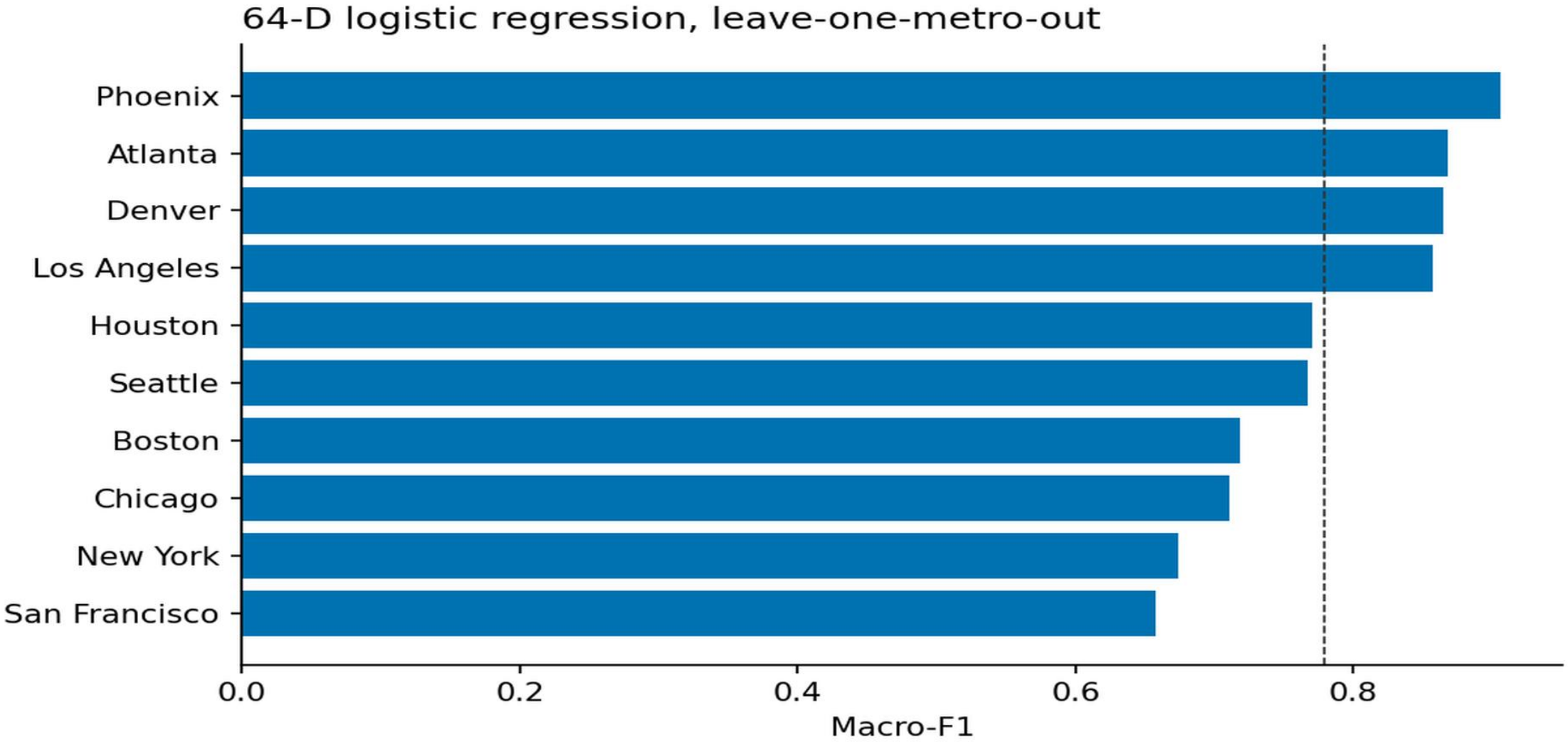
64-D latent

Frozen AlphaEarth

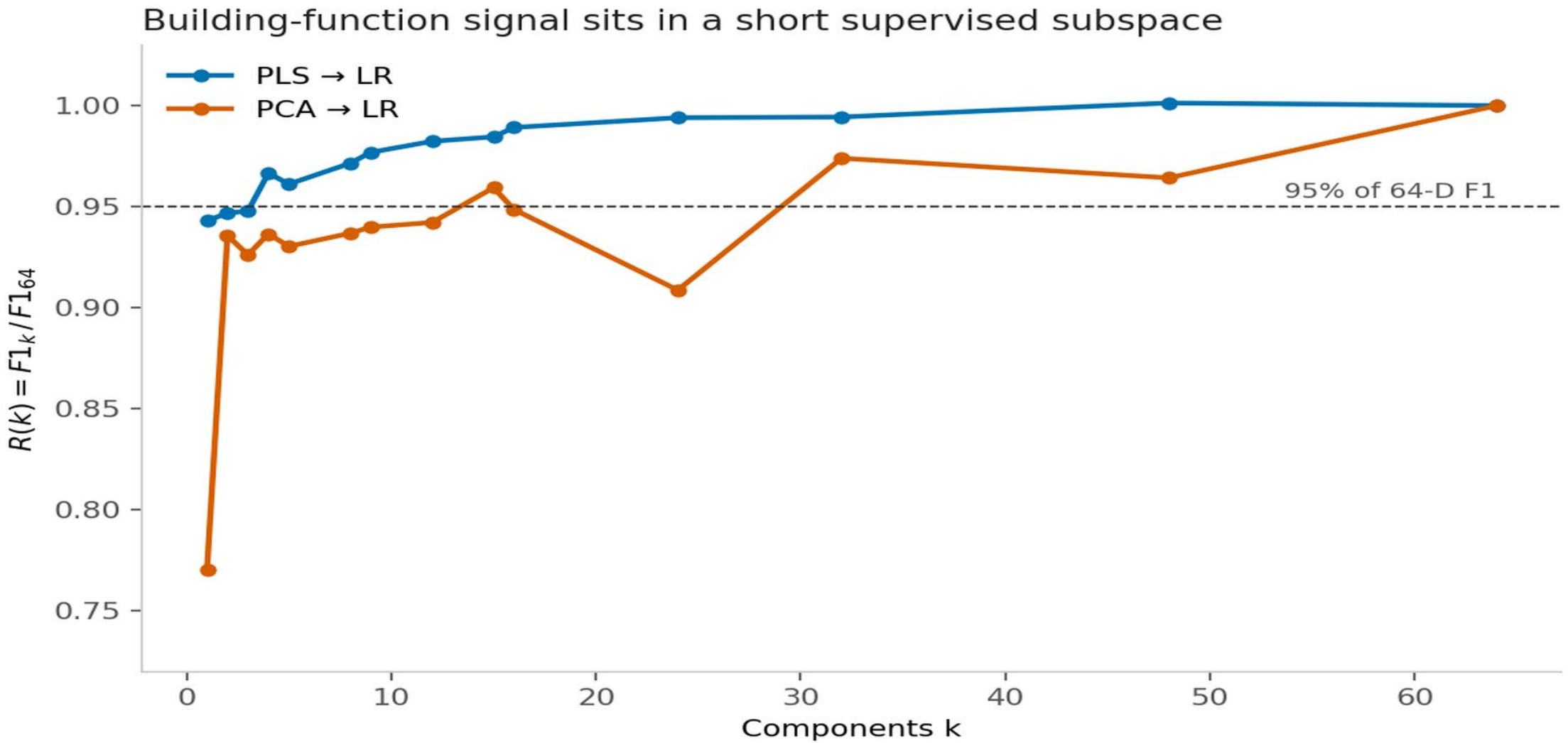
LORO split

Hold-out; threshold 0.50

Mean LORO F1


64-D logistic regression, leave-one-metro-out
Phoenix
Atlanta
Denver
Los Angeles
Houston
Seattle
Boston
Chicago
New York
San Francisco
0.0
0.2
0.4
0.6
0.8
Macro-F1

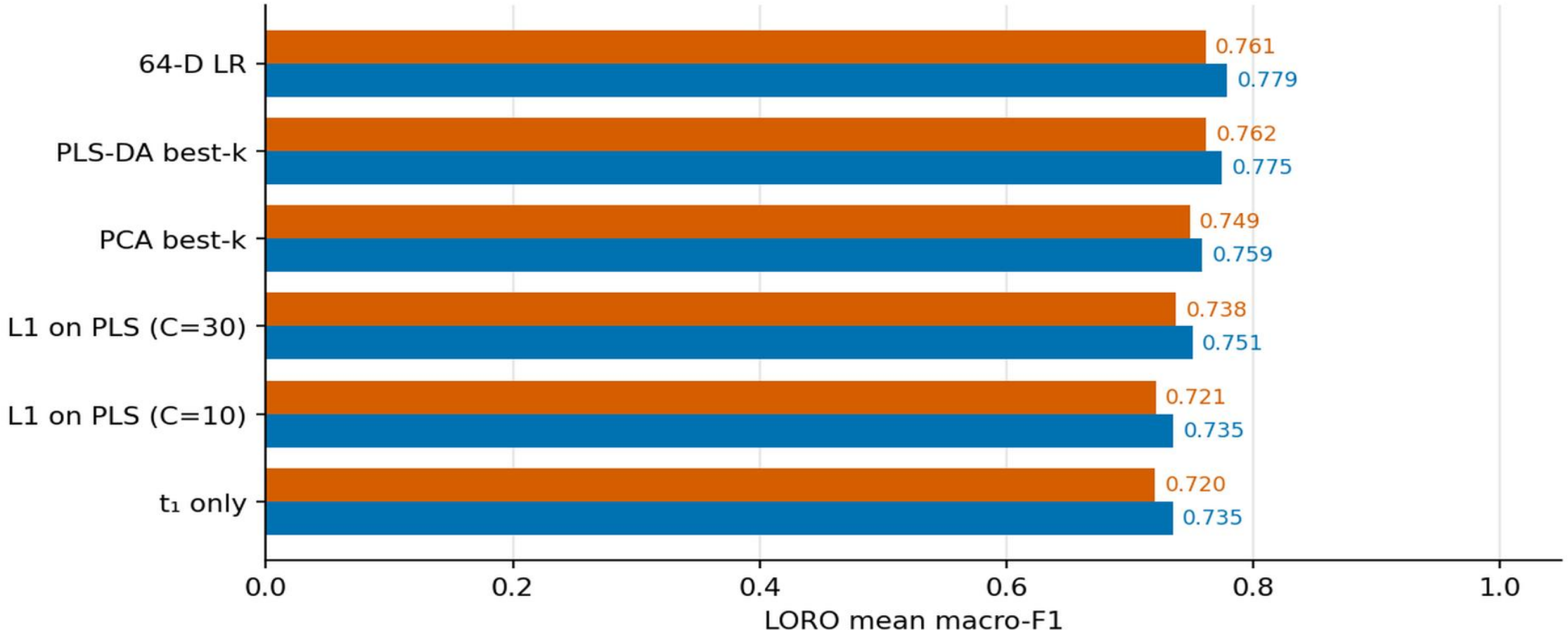
Building signal sits in a short supervised map



Mean-pooling to 100 m costs 0.01–0.02 F1

Every head is weaker at 100 m; t_1 still carries most of the signal

Blue = 10 m pure cells. Orange = 100 m mean-pooled AE. Same LORO metros.



Linear 100 m still wins on F1

Run	LORO F1	Mean gap
Mean-pool LR	0.761	—
ViT 128-D	0.756	—
v1 Supervised Contrastive (SupCon)	0.732	1.76
v3 longer	0.735	1.71
v4 gap loss	0.738	2.13

SECTION

ViT 3D tile experiment

Training pipeline · loss functions · what Supervised Contrastive (SupCon) did and did not do

ViT · TRAINING PIPELINE
100 m tile → 3D residential embedding

Leave-one-metro-out · each fold owns its own 3D axes

TRAINING PIPELINE

100 m tile → 3D residential embedding (leave-one-metro-out)



LORO: train on 9 metros · hold out 1 · each fold has its own 3D axes.

Batch (v4): one metro, balanced RES/NRES · Stop on validation sep/within (target 3).

ViT · LOSS FUNCTIONS
Two objectives tried on the same 3D vector

Left: BCE + Supervised Contrastive / SupCon (v1–v3; stop-grad on BCE head in v2/v3) · Right: BCE + pull + gap (v4)

v1–v3 · SUPERVISED CONTRASTIVE (SupCon)

BCE + λ · SupCon on z

$$\mathcal{L} = \mathrm{BCE}(s \cdot \cos(\hat{z}_{stop}, \hat{w}), y)$$
$$+\lambda\, \tau\, \mathcal{L}_{SupCon}(z, y)$$

stop-grad → BCE head (v2/v3)

v1: λ=1, τ=0.1 — BCE trains z (no stop-grad)
v2/v3: BCE head detached via stop-grad;
only SupCon trains the encoder
(λ=10, τ annealed 0.5 → 0.1)

Finding: with the head detached, val BCE fell but SupCon drifted back to chance — the two losses were not training the same weights.

v1 F1 0.732 · gap 1.76 | v2 0.632 · 2.12 | v3 0.735 · 1.71

v4 · WITHIN-REGION CLUSTER LOSS

BCE + pull + gap on the same z

$$\mathcal{L} = \mathrm{BCE}(s \cdot (z \cdot \hat{w}), y)$$
$$+\mathrm{mean}\,\|z - c_y\|^2$$
$$+\mathrm{relu}(3 - \mathrm{sep/within})^2$$

pull: tighten each class around its center c_Y
gap: push means apart until sep/within ≥ 3
Both terms act on the same embedding z
(no stop-grad — BCE and cluster share z).

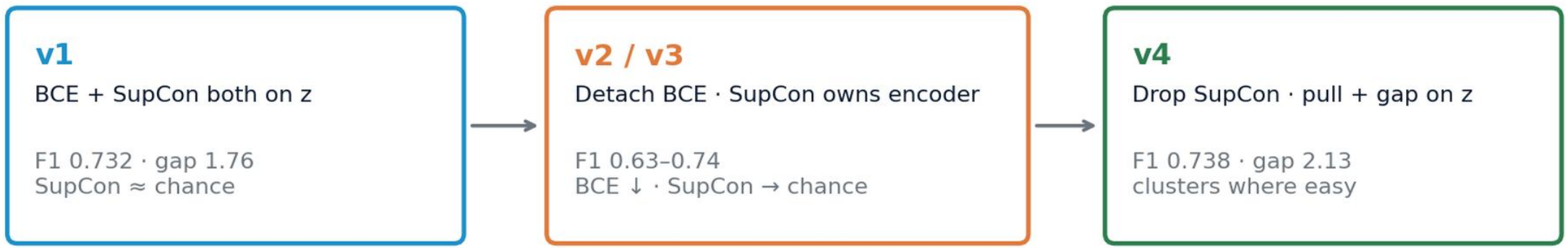
Finding: mean held-out gap → 2.13; LORO F1 0.738 (still below mean-pool LR 0.761). Atlanta/Phoenix clear; NYC gap 0.63.
Stop: mean val sep/within · min 35 ep · patience 6 · cap 60

ViT · CONTRASTIVE FINDINGS
Supervised Contrastive (SupCon) did not deliver separable held-out clusters

Detach the classifier and the encoder drifts back toward chance

WHAT THE CONTRASTIVE EXPERIMENTS SHOWED

Supervised Contrastive (SupCon) did not deliver separable 3D clusters under LORO



Takeaway: a Supervised Contrastive term that does not share gradients with the classifier does not make held-out metros separate. Even with explicit gap loss (v4), linear mean-pool still wins on LORO F1 (0.761 vs 0.738).

Ladder vs linear 100 m mean-pool

Raising the held-out gap did not beat linear F1

RESULTS

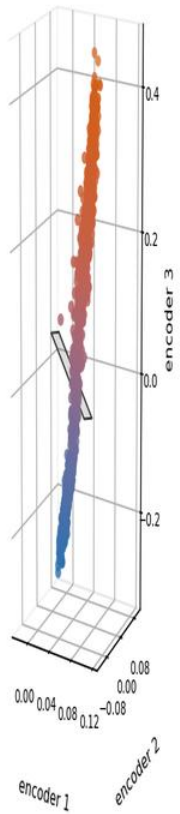
ViT 3D ladder vs linear 100 m

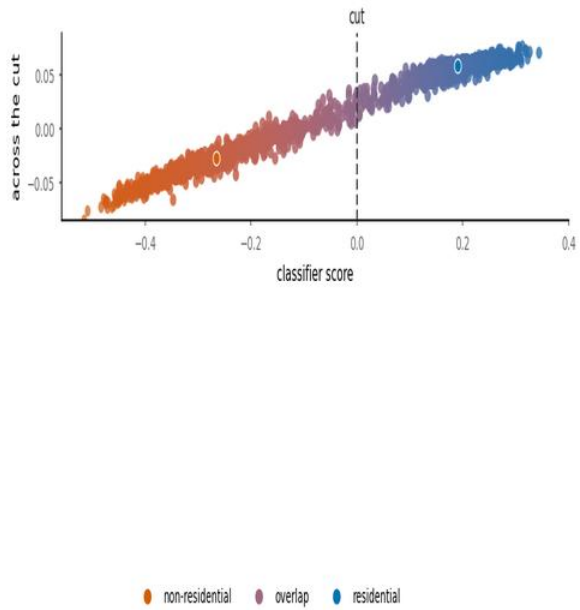
Run	LORO macro F1	Mean held-out gap
100 m mean-pool LR	**0.761**	**—**
ViT 128-D class token	0.756	—
v1 BCE + Supervised Contrastive (SupCon)	0.732	1.76
v2 detached head + SupCon	0.632	2.12
v3 longer train, stop on loss	0.735	1.71
v4 pull + gap (target 3)	0.738	2.13

Atlanta clears a gap of 3.5 · NYC does not

Atlanta: residential and non-residential are apart

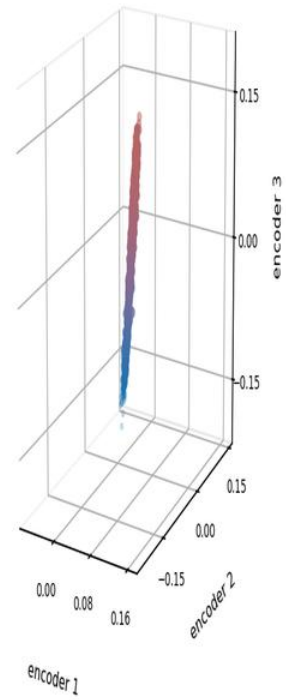
Held-out tiles. F1 0.898. Gap 3.51. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.

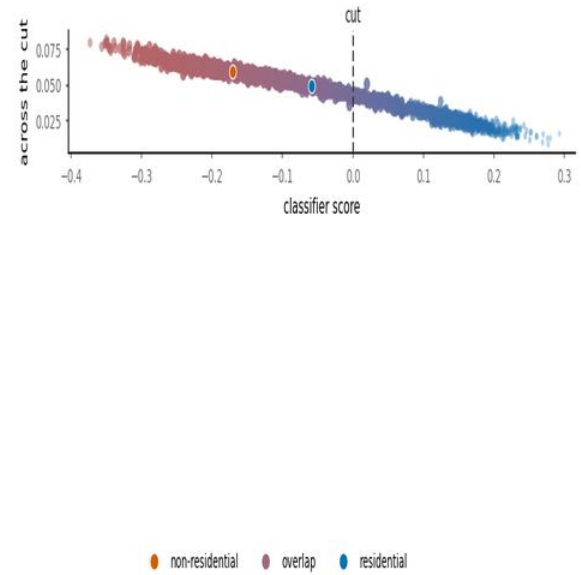




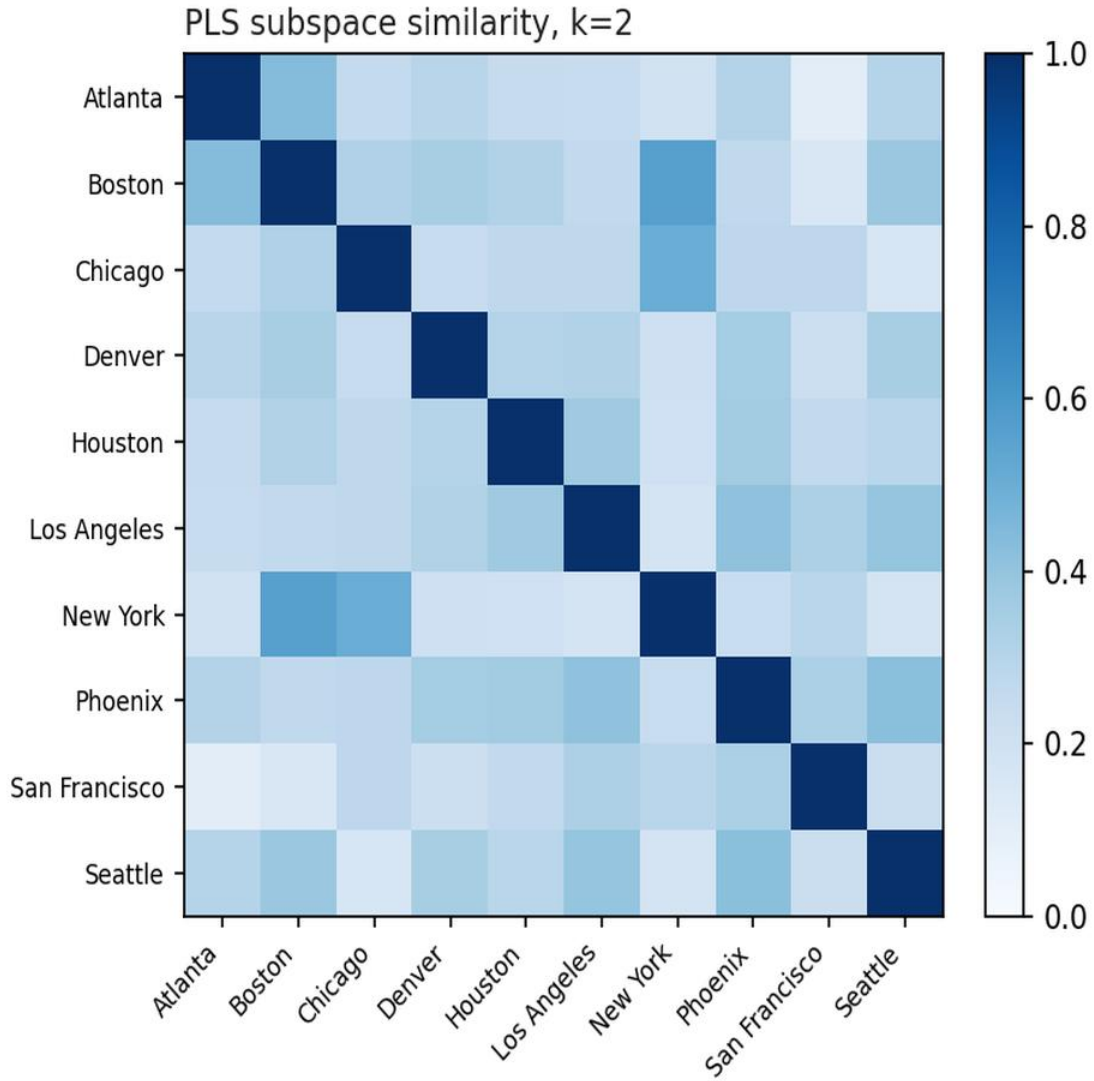
New York: residential and non-residential are one cloud

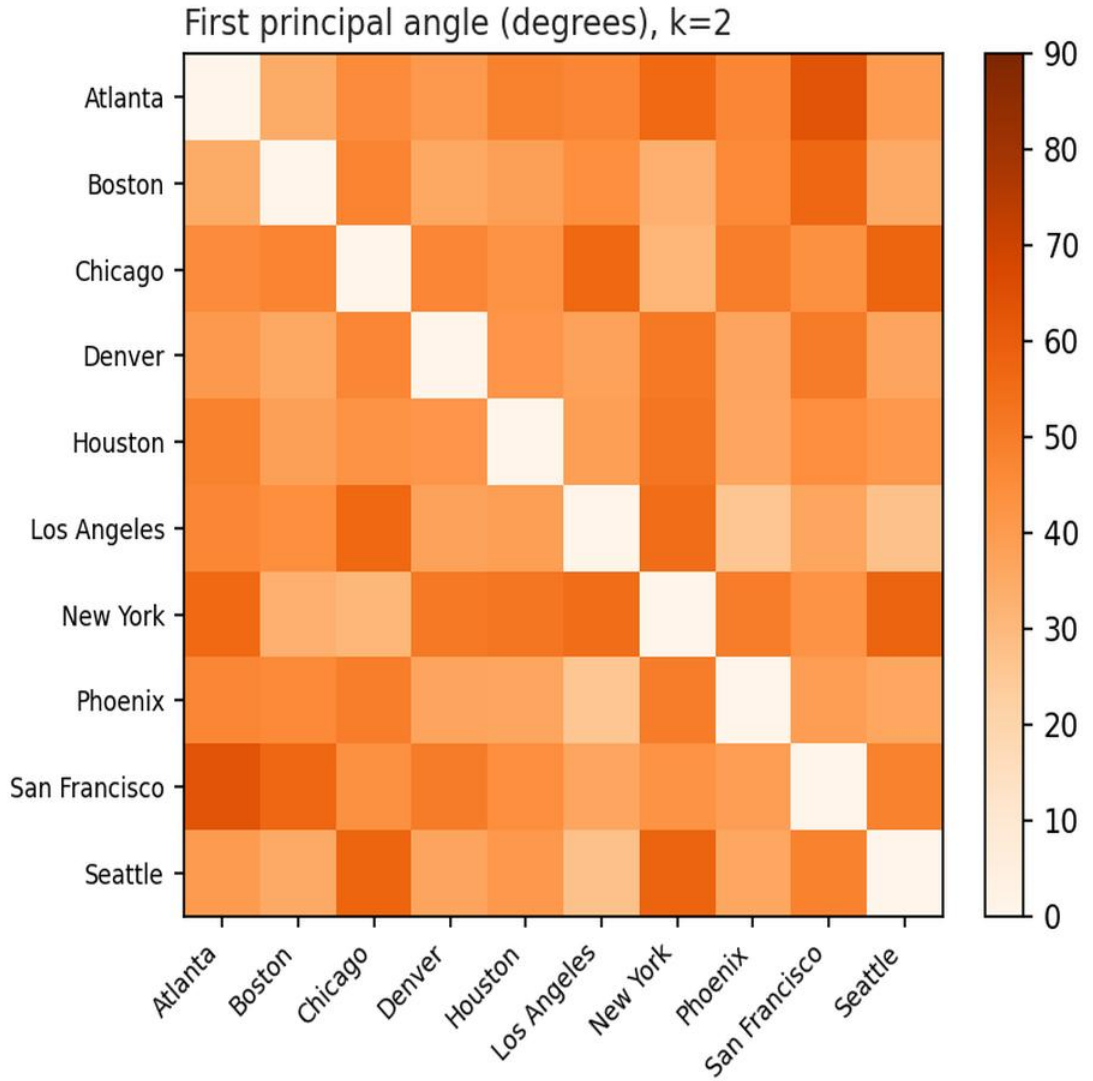
Held-out tiles. F1 0.586. Gap 0.63. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



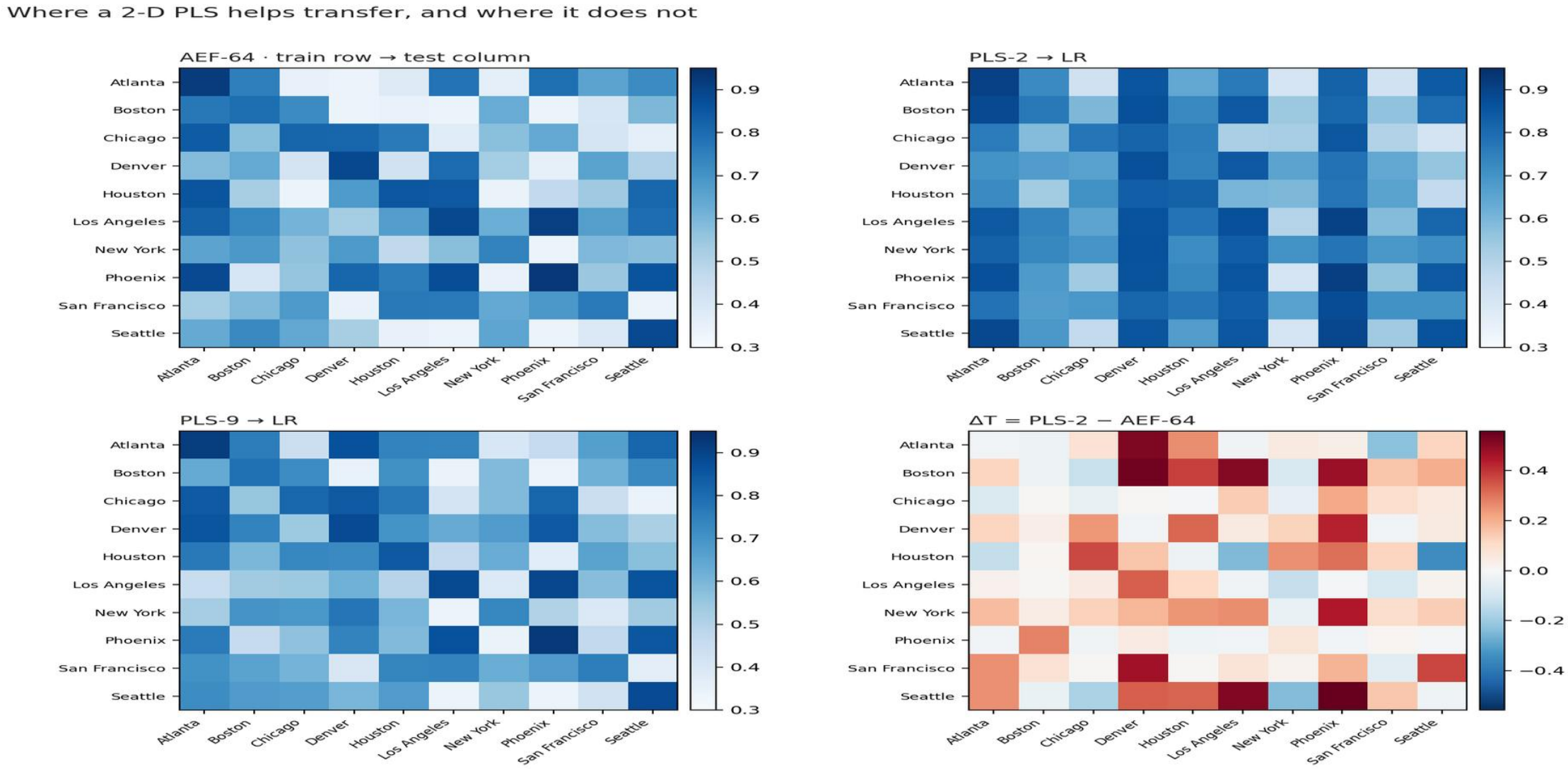


Geographic subspaces — three questions

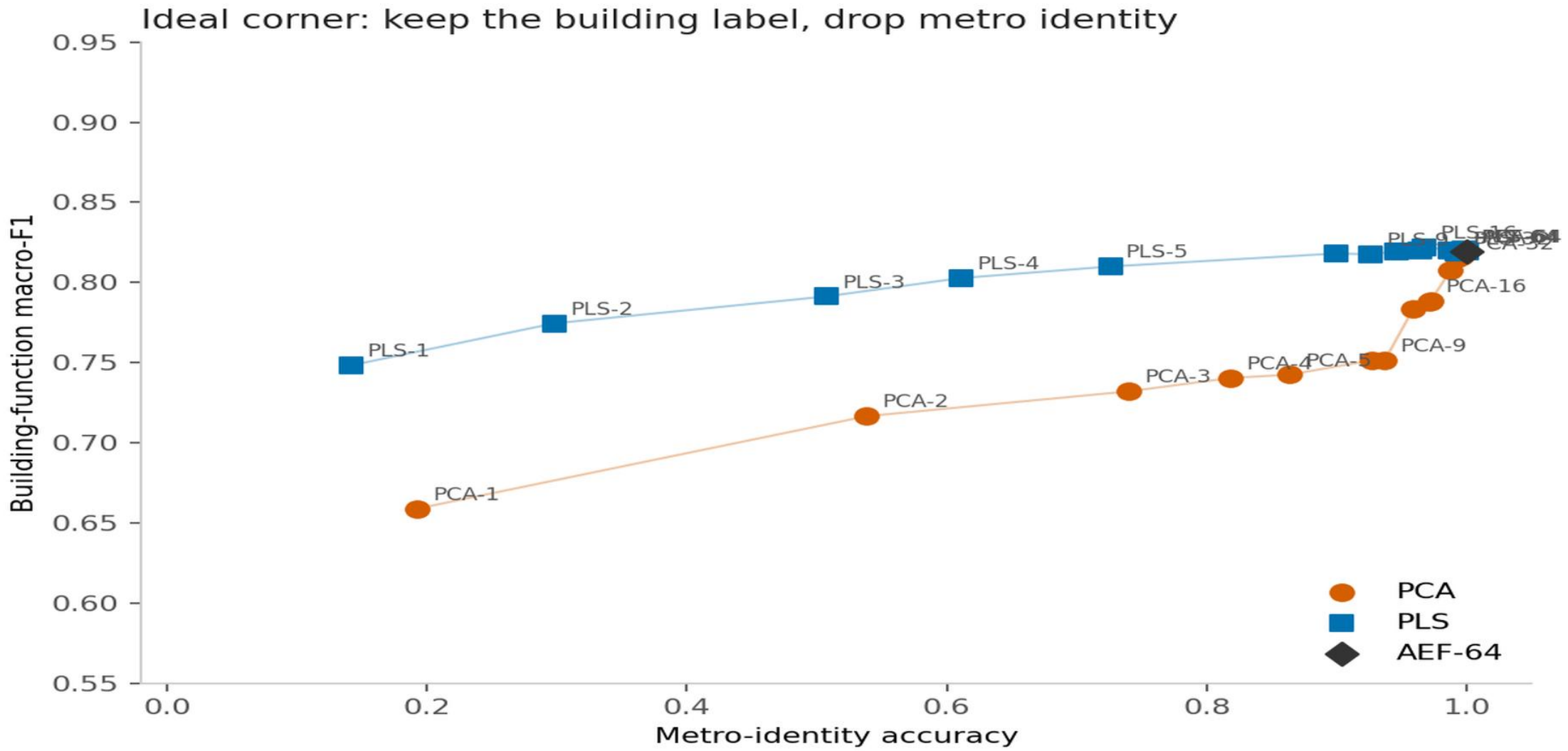




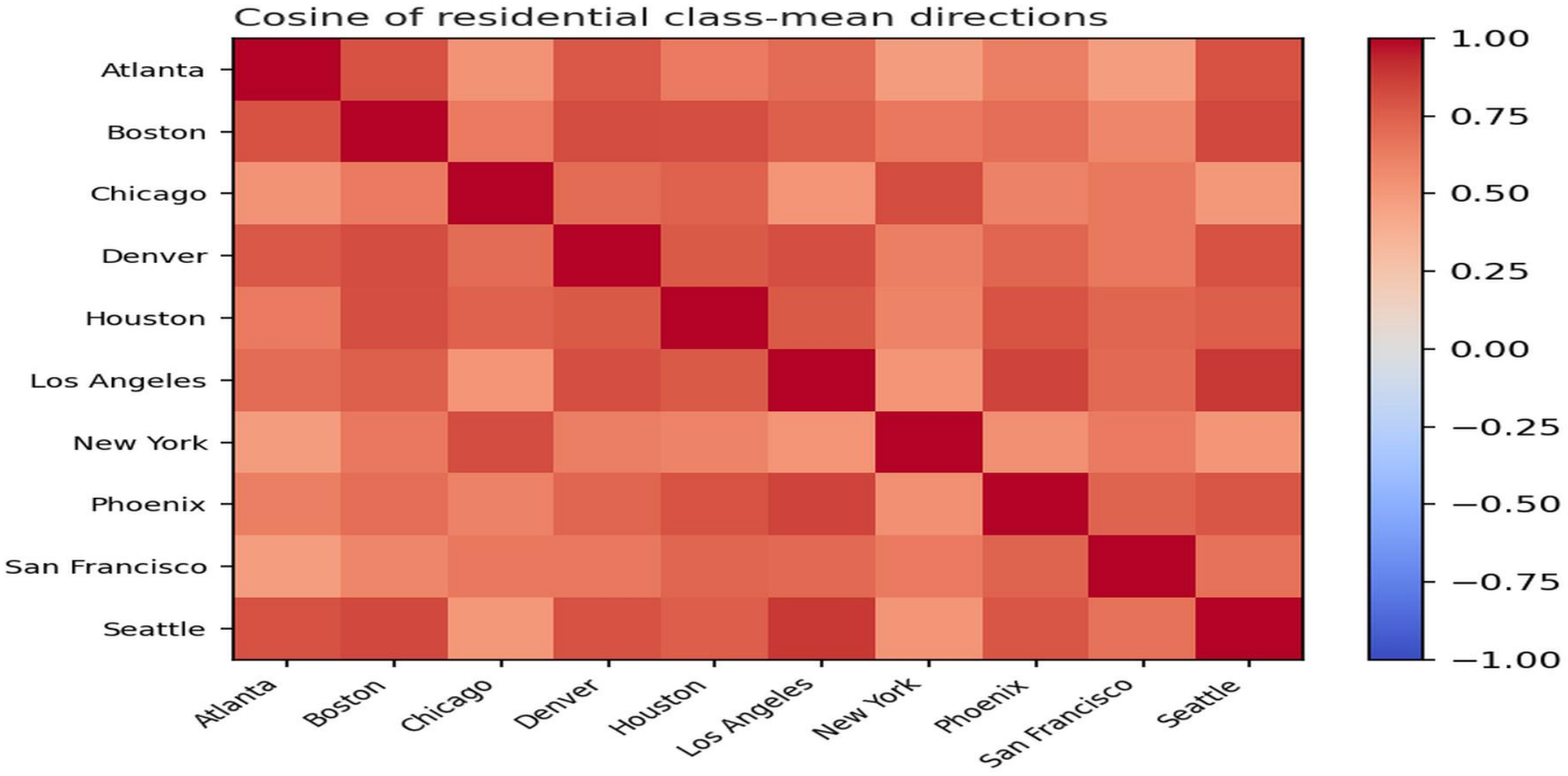
Short PLS transfers better than full 64-D



PLS k=2 keeps the label · drops metro identity



NYC aligns with Chicago — not easy targets



What holds · what does not

~0.78 LORO F1 in frozen AE

Short PLS carries most of it

Subspaces only partly aligned

PLS k=2 ships better than 64-D

Alignment causes transfer

Global geographic generalization

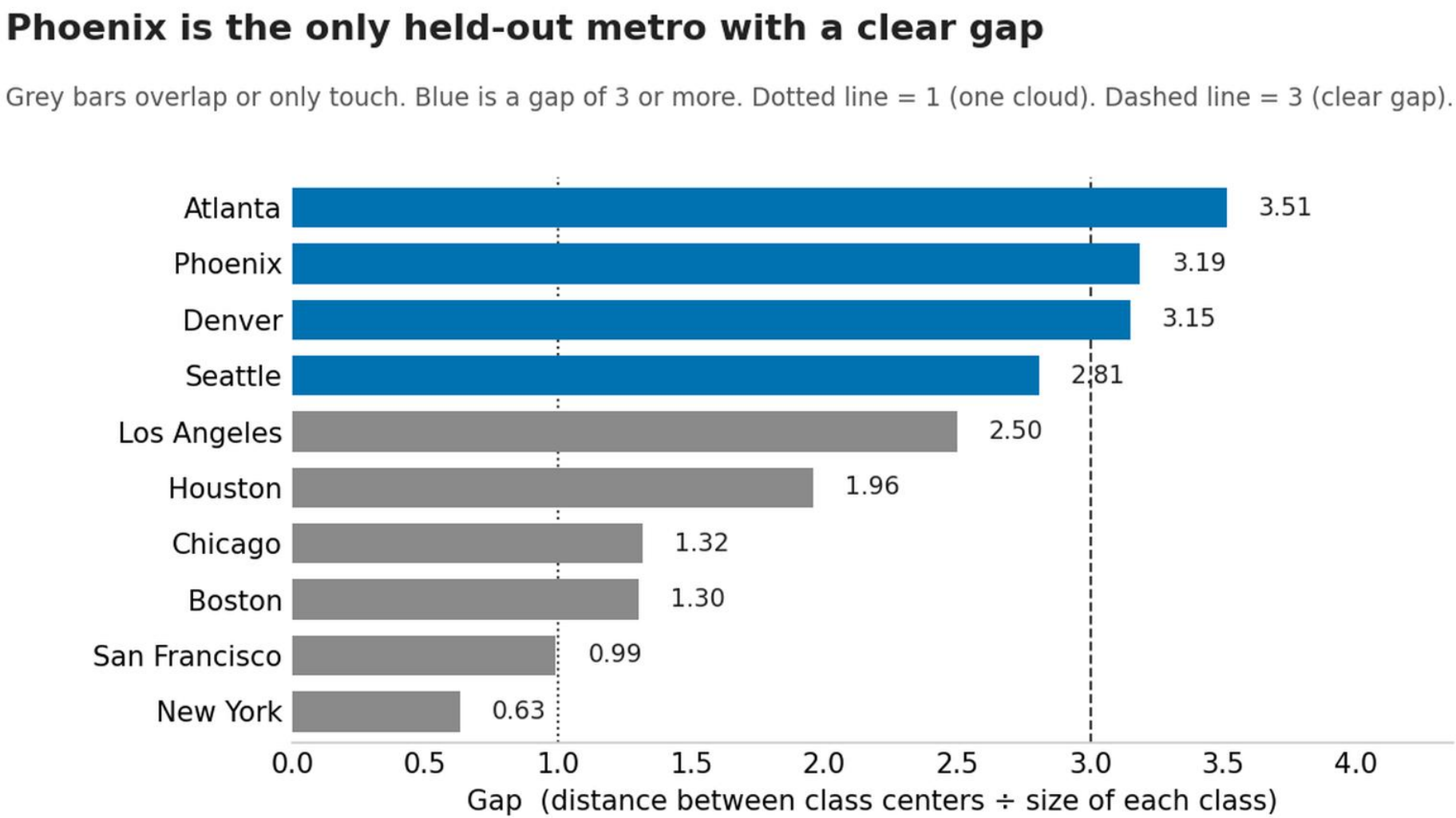
v4 fails because the axis is wrong

Appendix · v4 LORO held-out clusters

All ten metros · largest gap first

APPENDIX · CLUSTER
Held-out gap by metro

overview

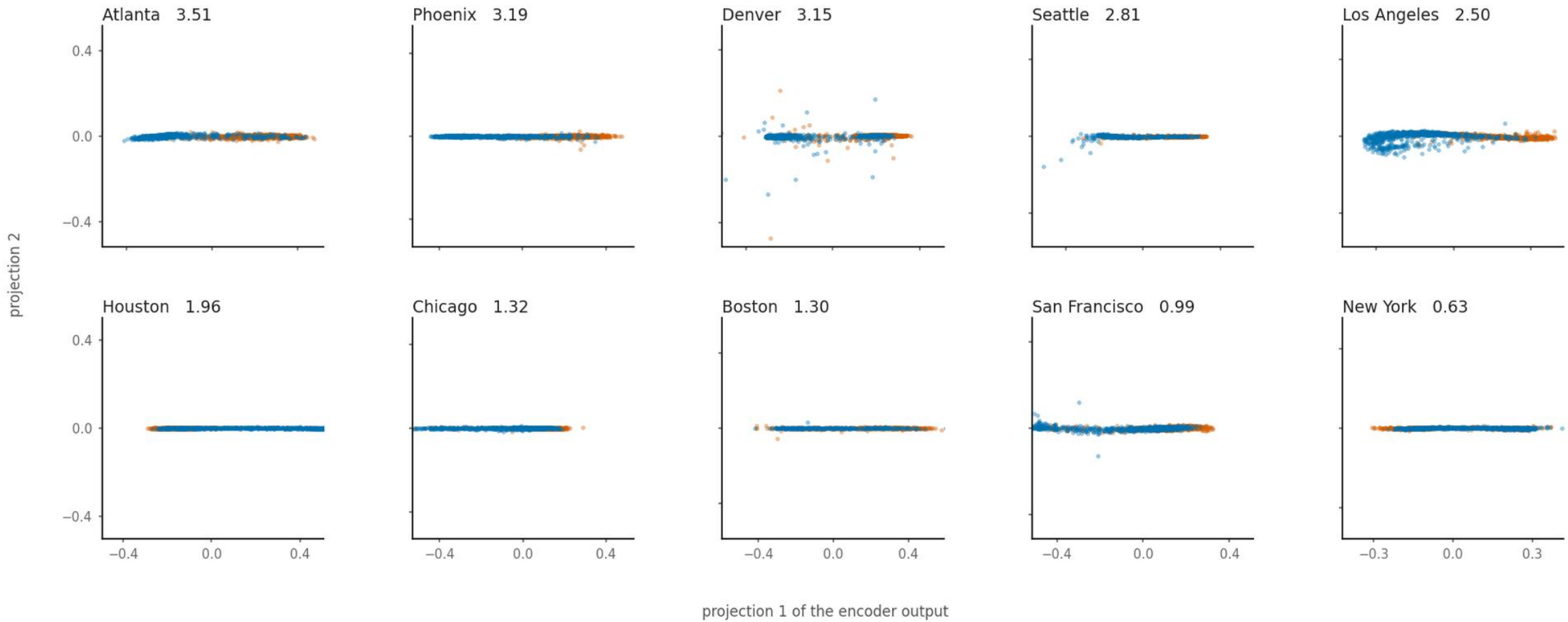


APPENDIX · CLUSTER
2D projection

overview

The classes pull apart in Phoenix and sit together in New York

Each panel projects that metro's three encoder outputs onto the plane they spread across. Own scale. The number is the gap. Blue is residential. Orange is non-residential.

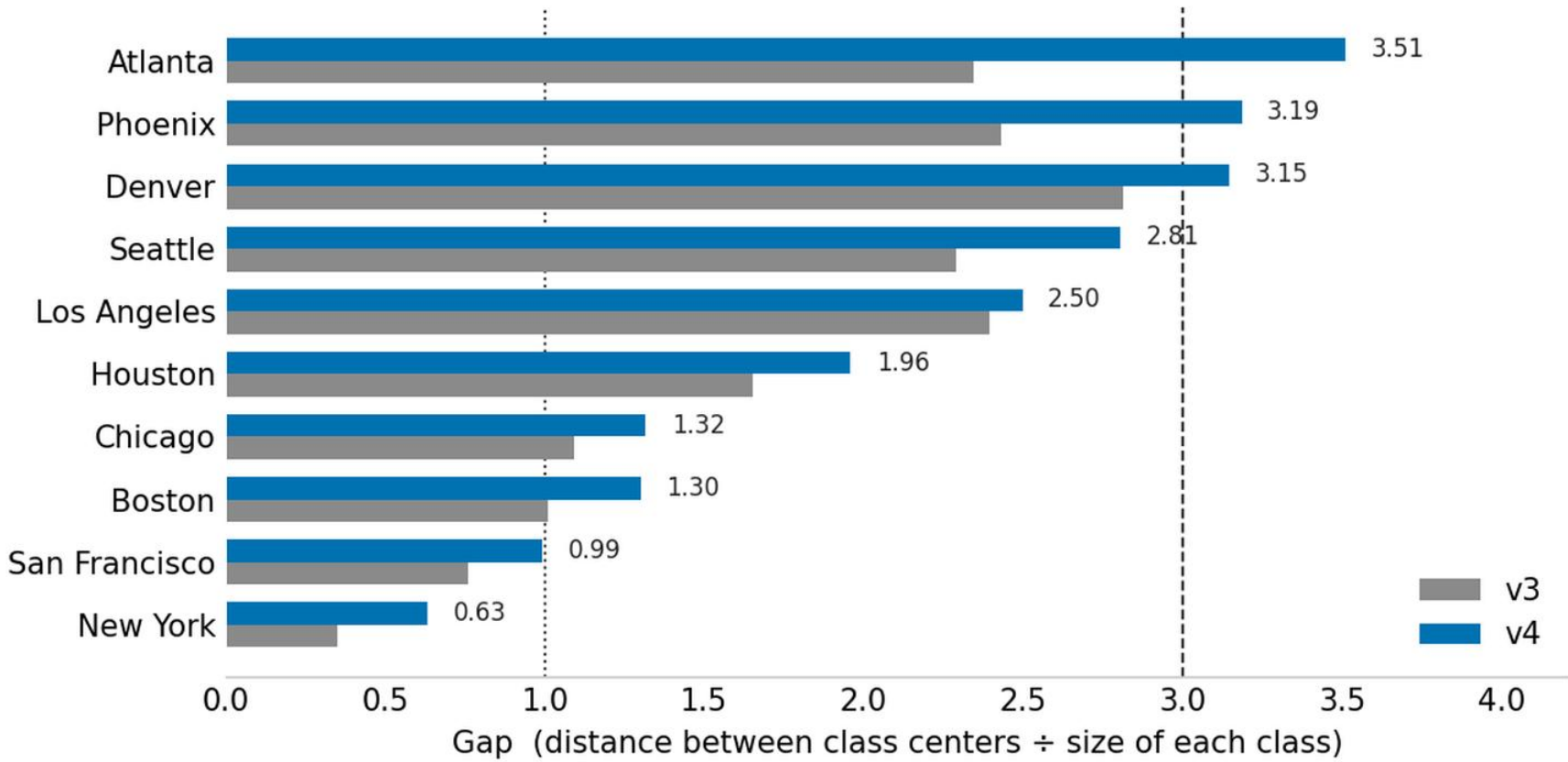


APPENDIX · CLUSTER
v4 vs v3 gap

overview

Atlanta, Phoenix, and Denver clear a gap of 3

Held-out 100 m tiles. Blue is v4. Grey is v3. Dotted line = 1 (one cloud). Dashed line = 3 (clear gap).

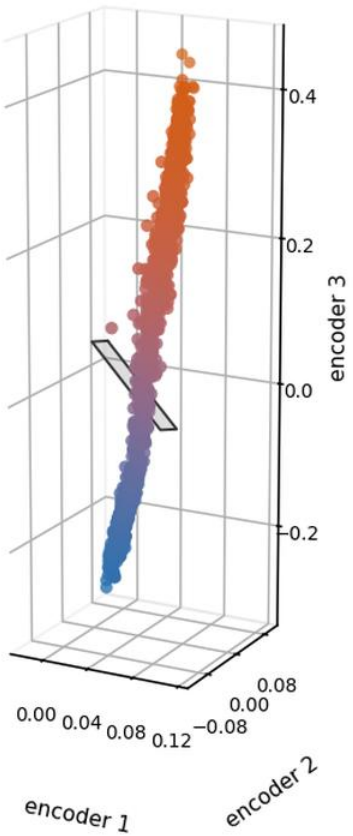


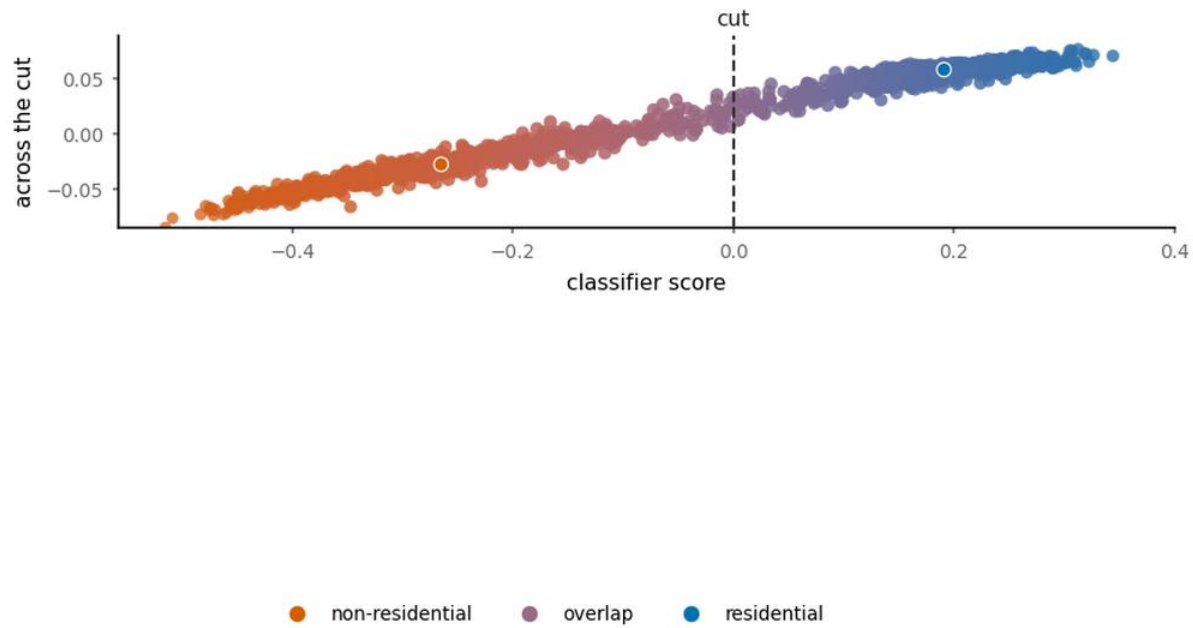
APPENDIX · CLUSTER
LORO held-out · Atlanta SE

F1 0.898 · gap 3.51

Atlanta: residential and non-residential are apart

Held-out tiles. F1 0.898. Gap 3.51. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



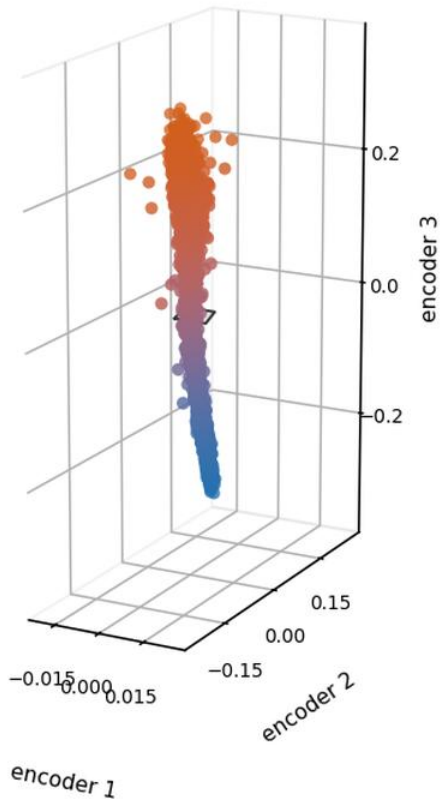


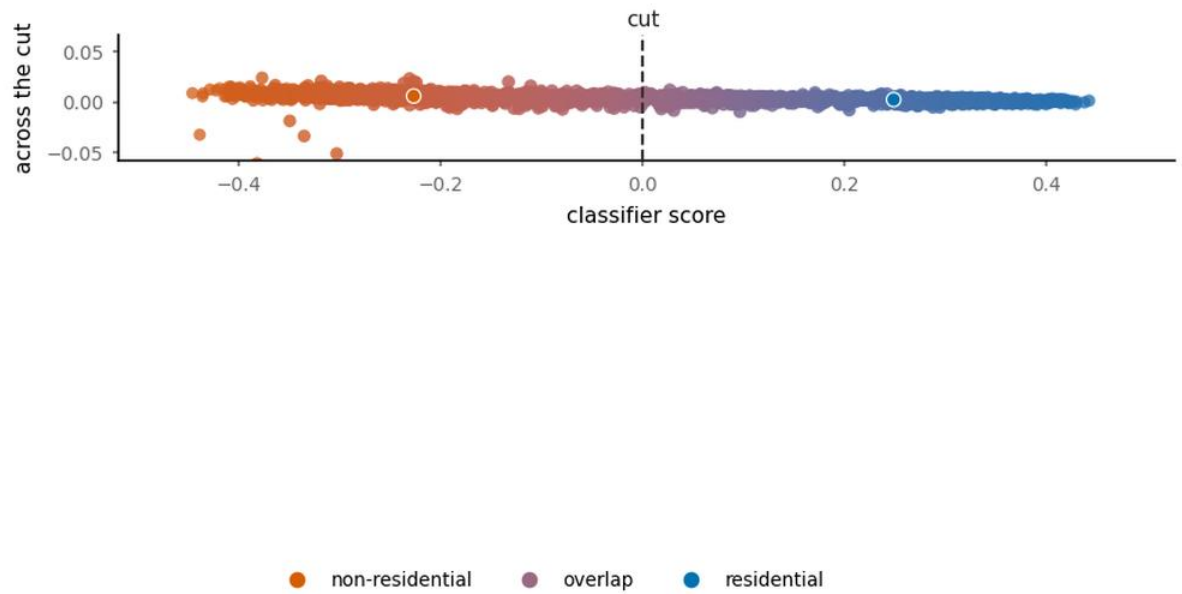
APPENDIX · CLUSTER
LORO held-out · Phoenix SW

F1 0.878 · gap 3.19

Phoenix: residential and non-residential are apart

Held-out tiles. F1 0.878. Gap 3.19. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



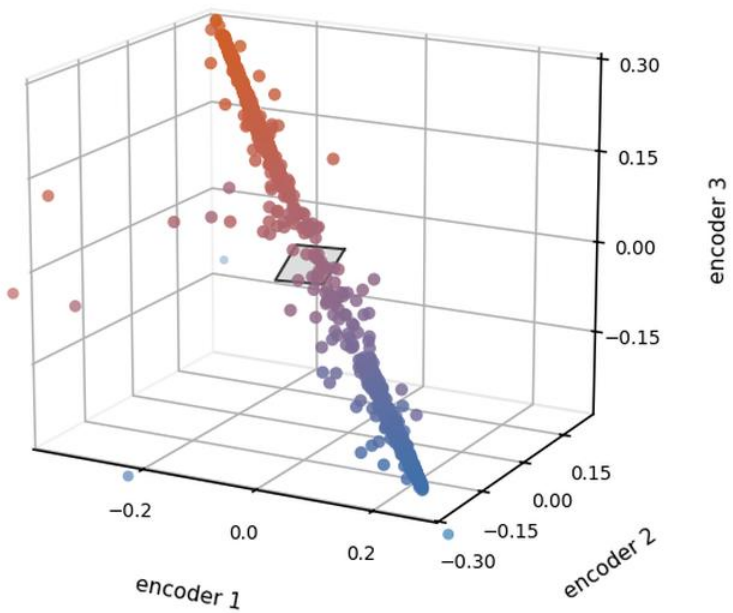


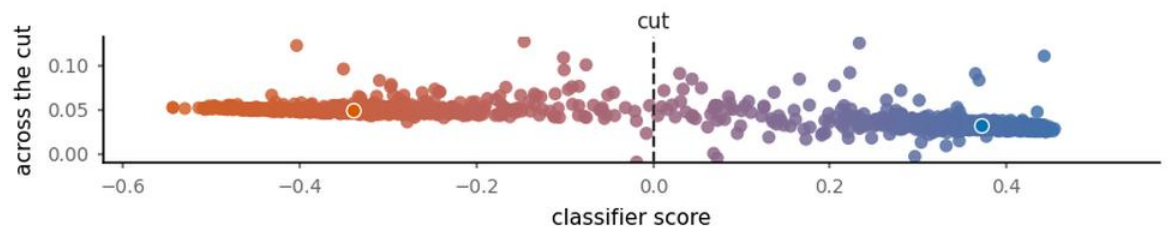
APPENDIX · CLUSTER
LORO held-out · Denver W

F1 0.862 · gap 3.15

Denver: residential and non-residential are apart

Held-out tiles. F1 0.862. Gap 3.15. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.





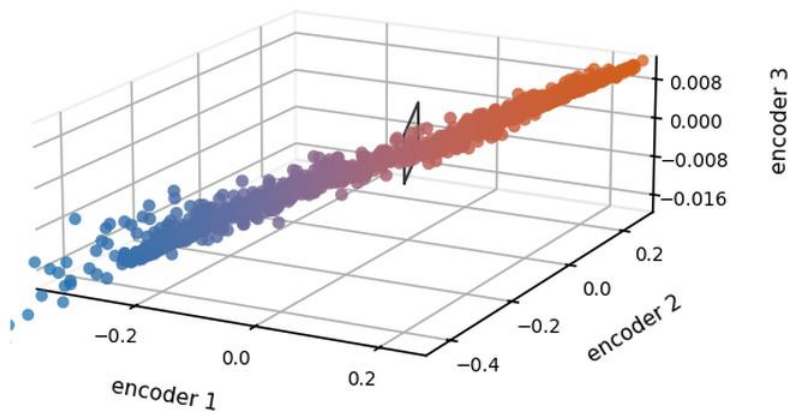
non-residential overlap residential

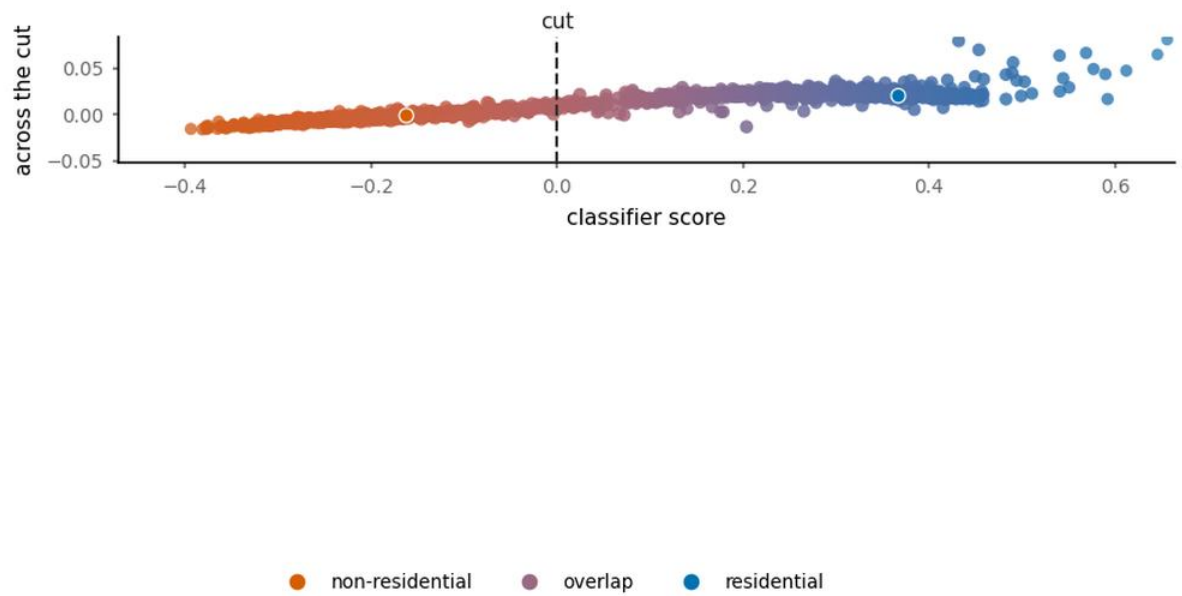
APPENDIX · CLUSTER
LORO held-out · Seattle NW

F1 0.824 · gap 2.81

Seattle: residential and non-residential are apart

Held-out tiles. F1 0.824. Gap 2.81. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



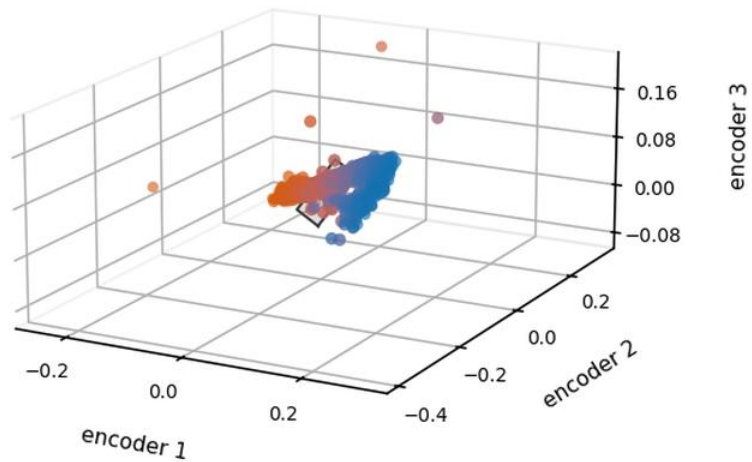


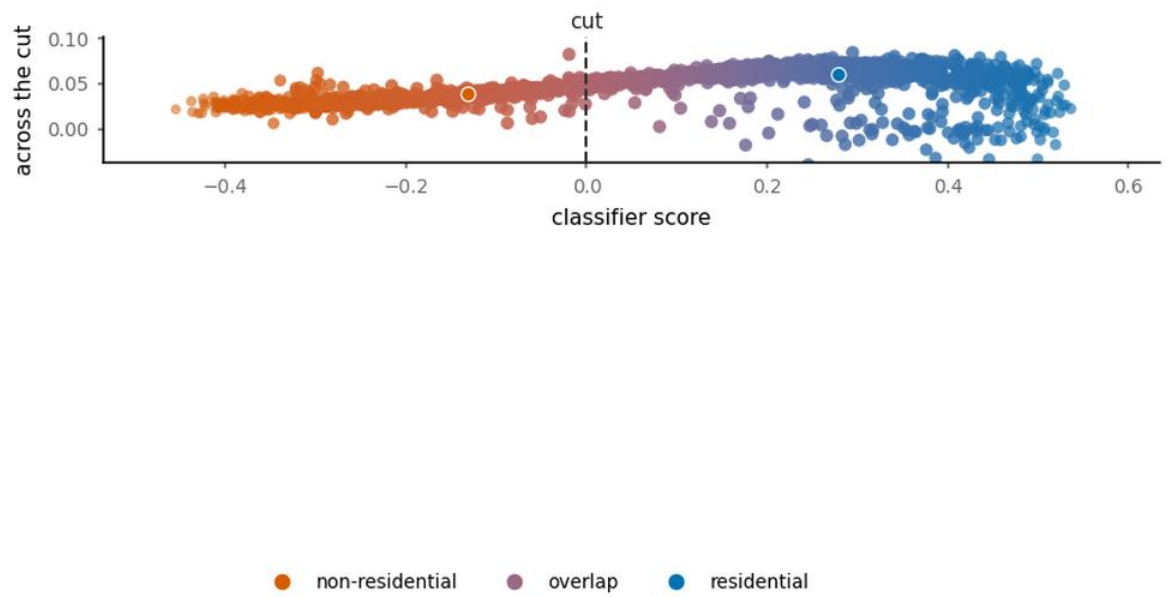
APPENDIX · CLUSTER
LORO held-out · LA W

F1 0.793 · gap 2.50

Los Angeles: the two classes touch

Held-out tiles. F1 0.793. Gap 2.50. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



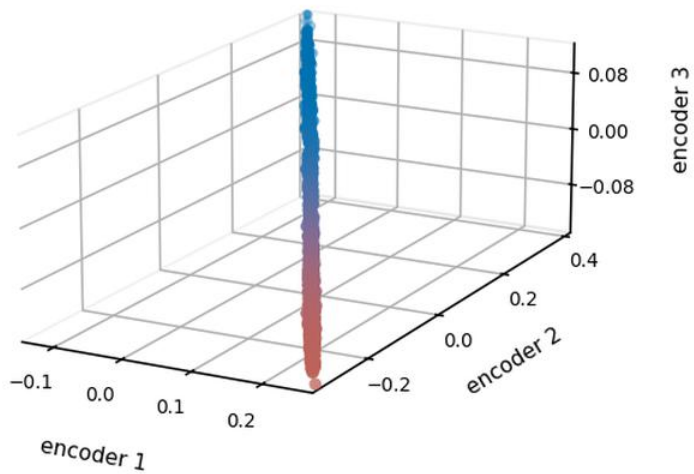


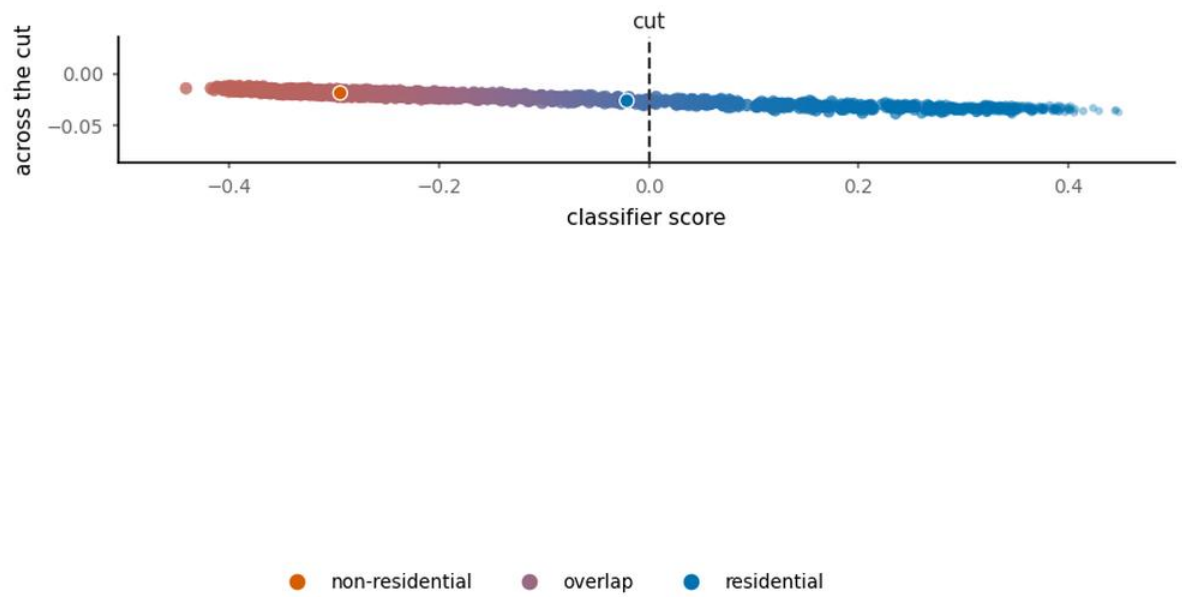
APPENDIX · CLUSTER
LORO held-out · Houston S

F1 0.709 · gap 1.96

Houston: the two classes touch

Held-out tiles. F1 0.709. Gap 1.96. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



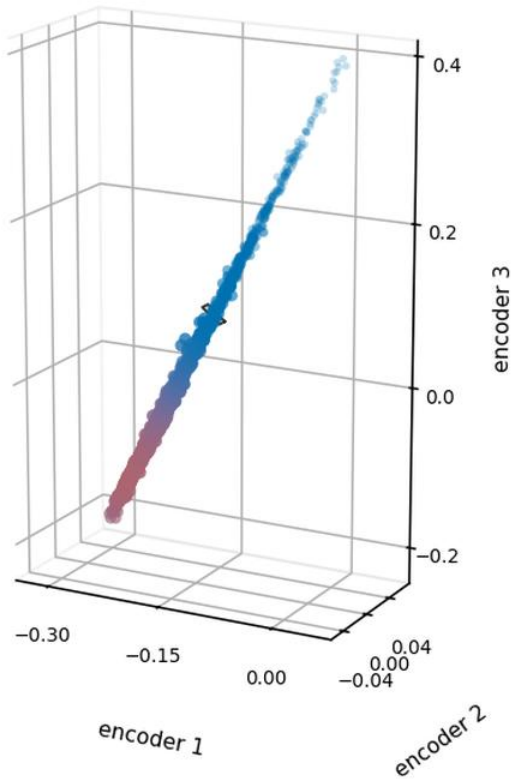


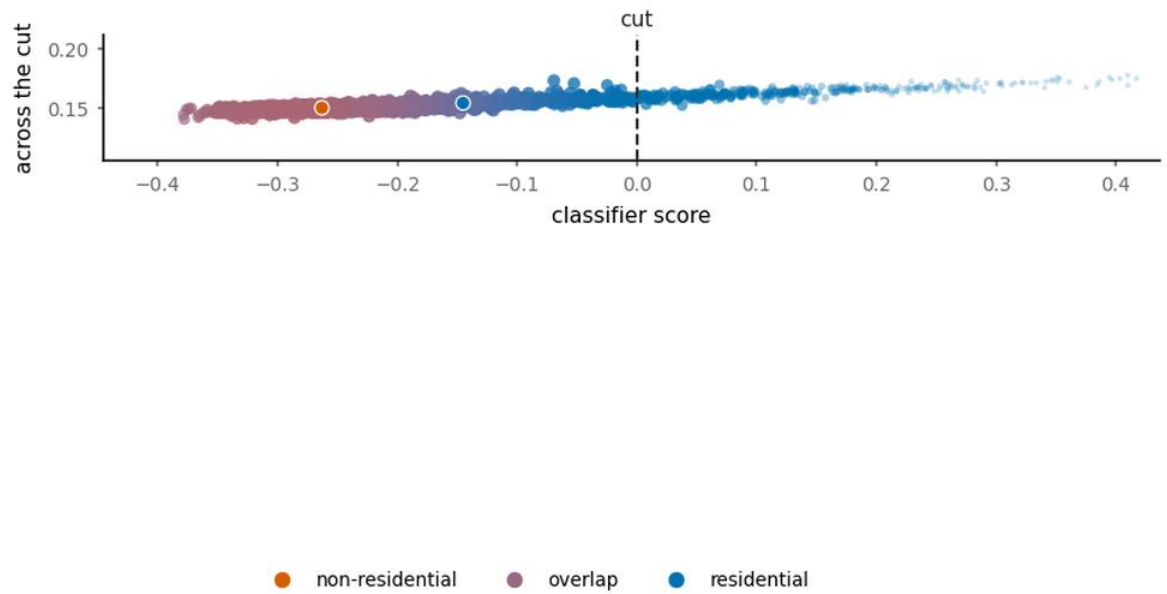
APPENDIX · CLUSTER
LORO held-out · Chicago MW

F1 0.553 · gap 1.32

Chicago: residential and non-residential are one cloud

Held-out tiles. F1 0.553. Gap 1.32. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



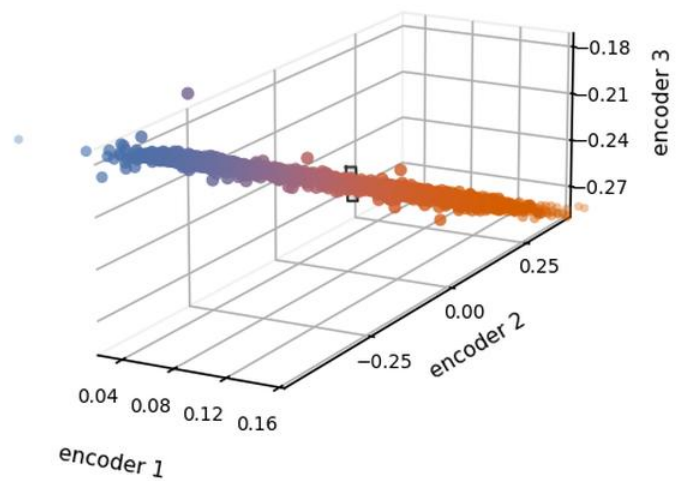


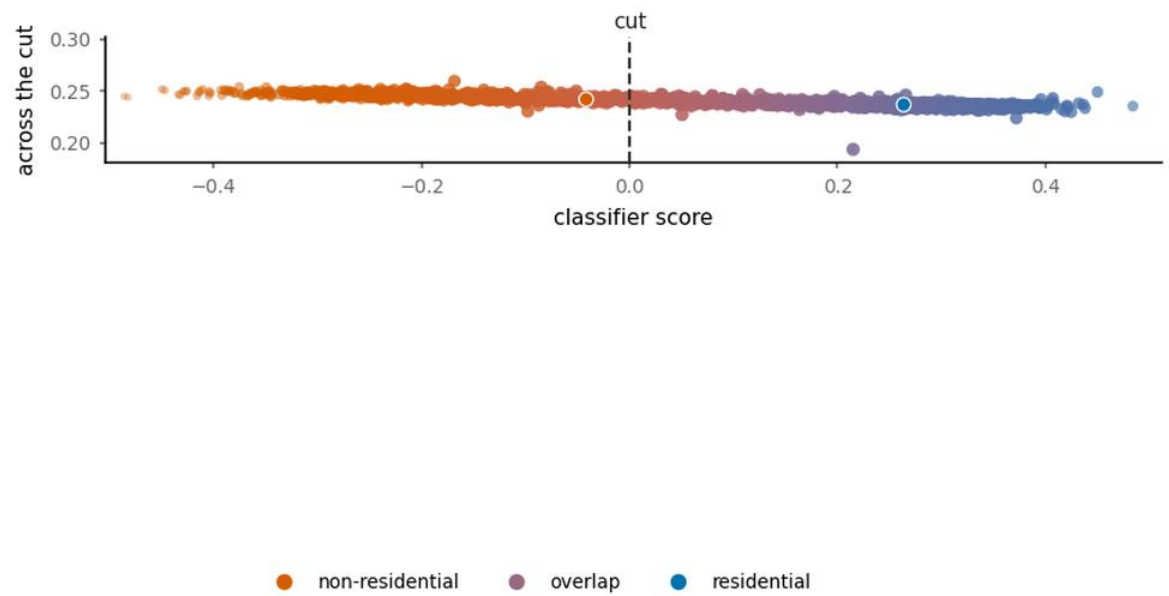
APPENDIX · CLUSTER
LORO held-out · Boston NE

F1 0.693 · gap 1.30

Boston: residential and non-residential are one cloud

Held-out tiles. F1 0.693. Gap 1.30. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



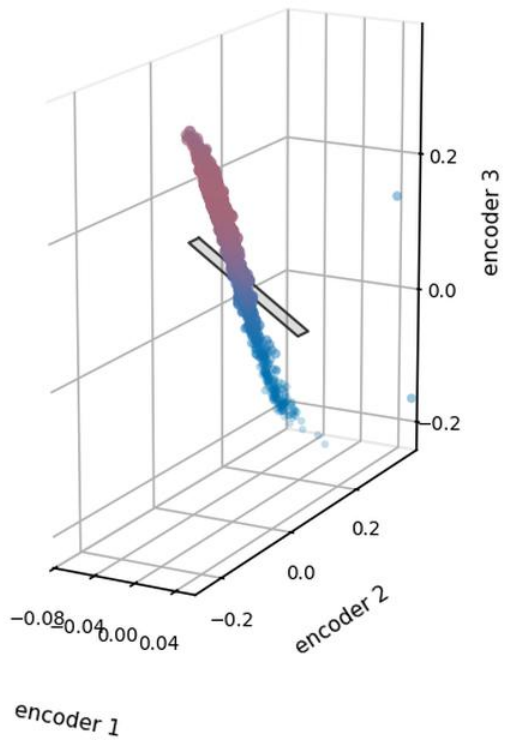


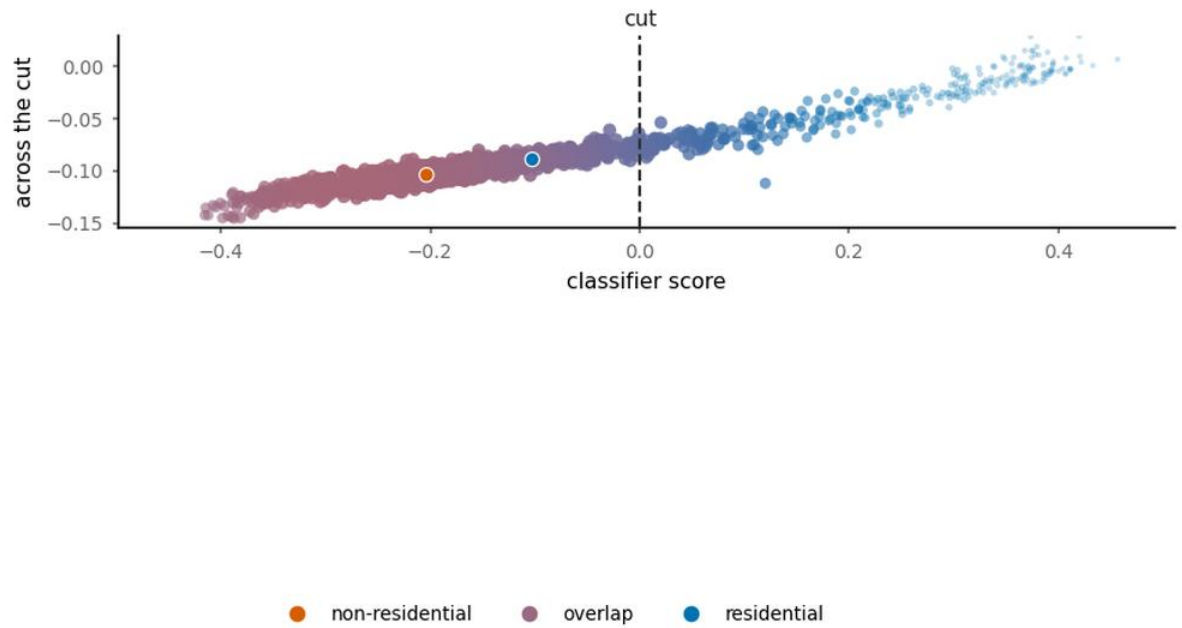
APPENDIX · CLUSTER
LORO held-out · SF W

F1 0.584 · gap 0.99

San Francisco: residential and non-residential are one cloud

Held-out tiles. F1 0.584. Gap 0.99. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.



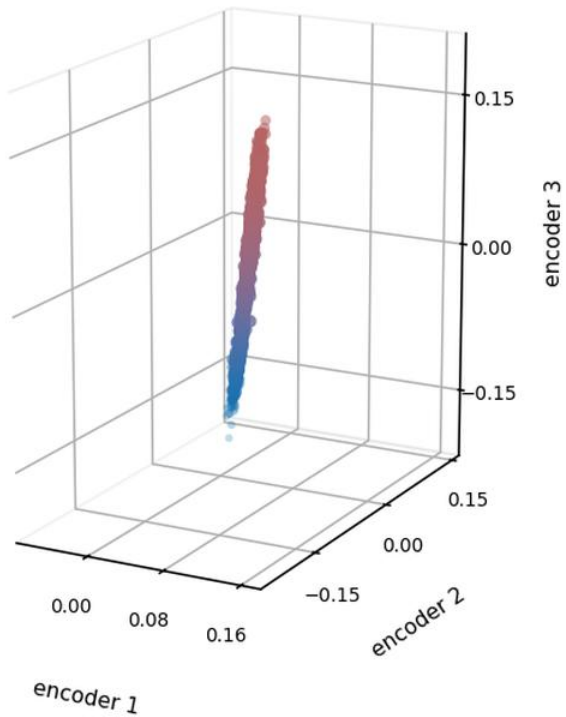


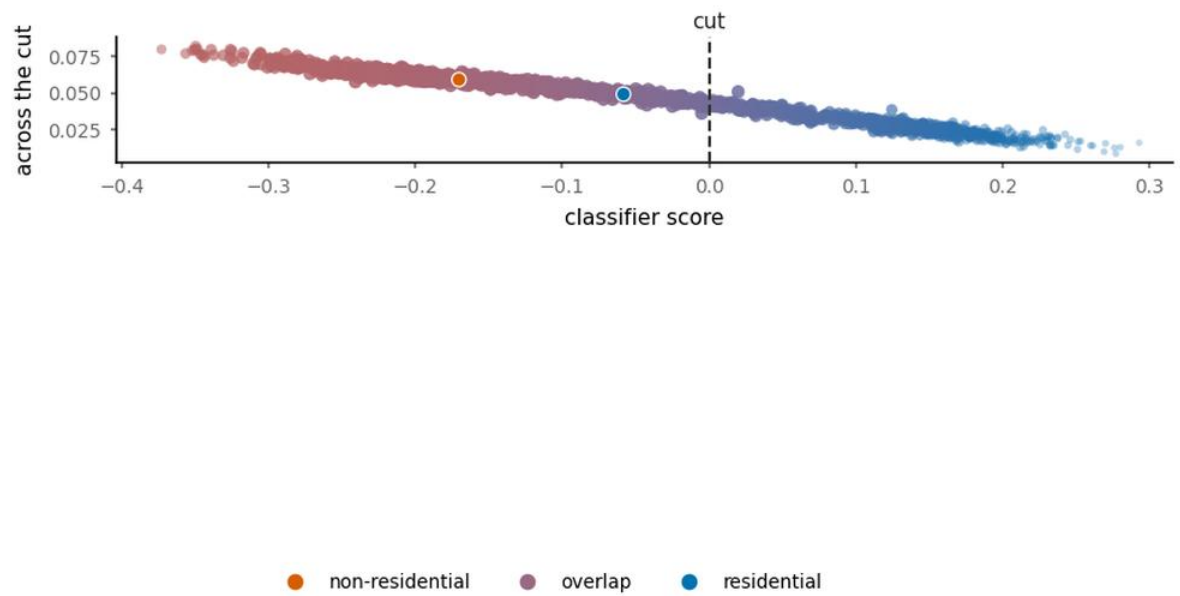
APPENDIX · CLUSTER
LORO held-out · NYC NE

F1 0.586 · gap 0.63

New York: residential and non-residential are one cloud

Held-out tiles. F1 0.586. Gap 0.63. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.





Appendix · validation while NYC held out

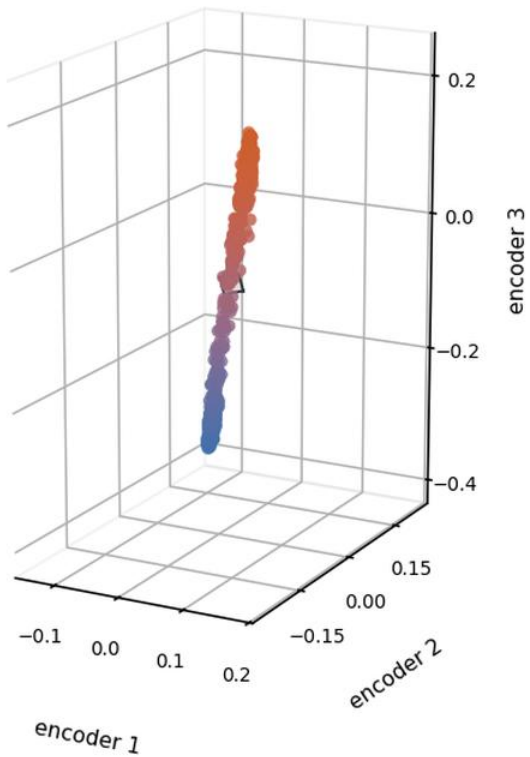
Same NYC-fold checkpoint

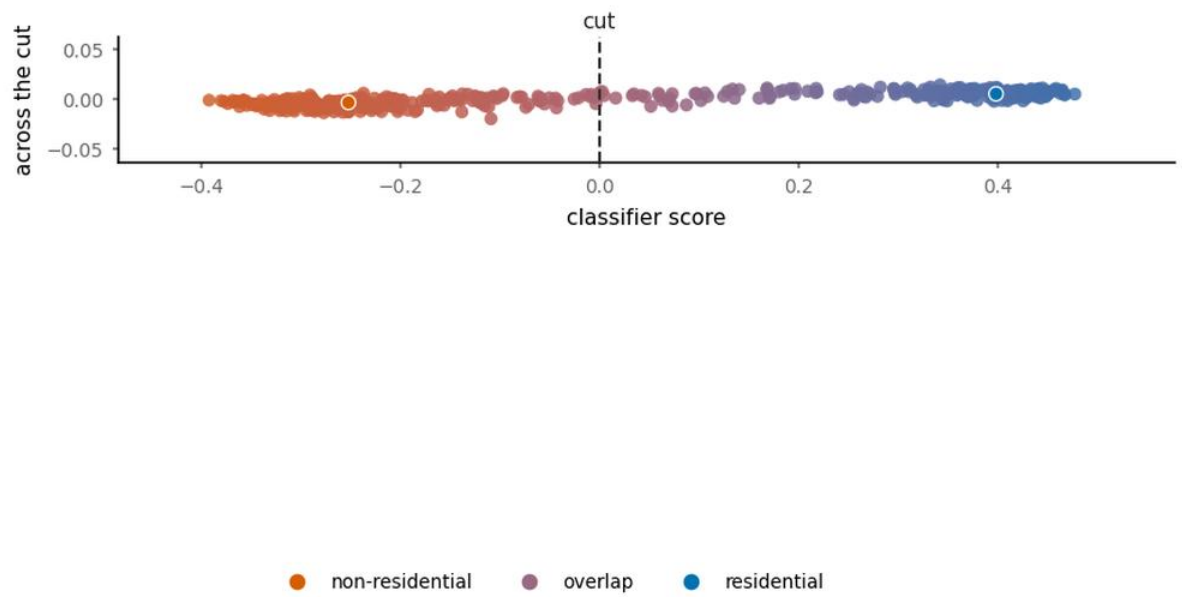
APPENDIX · CLUSTER
NYC-fold validation · Phoenix SW

Gap 5.44

Phoenix: residential and non-residential are apart

Validation tiles while New York is held out. n=514 (289 non-residential, 225 residential). Gap 5.44. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



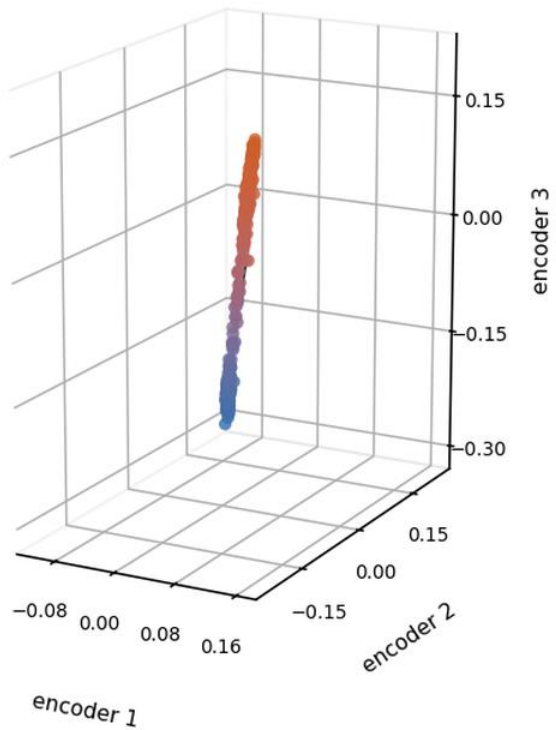


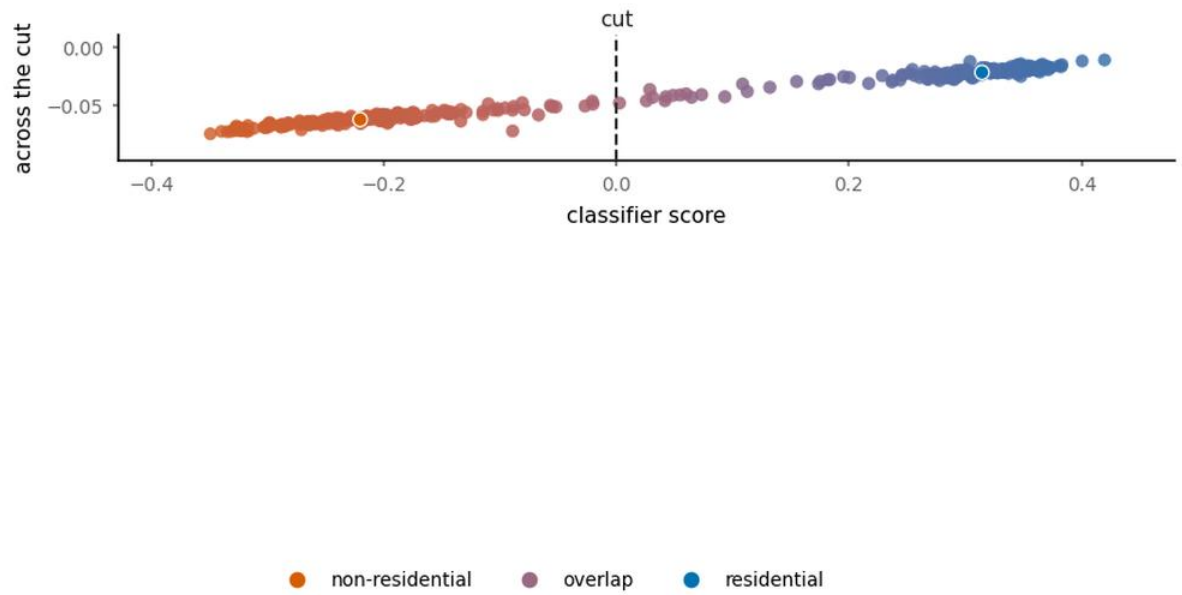
APPENDIX · CLUSTER
NYC-fold validation · Atlanta SE

Gap 4.98

Atlanta: residential and non-residential are apart

Validation tiles while New York is held out. n=367 (180 non-residential, 187 residential). Gap 4.98. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



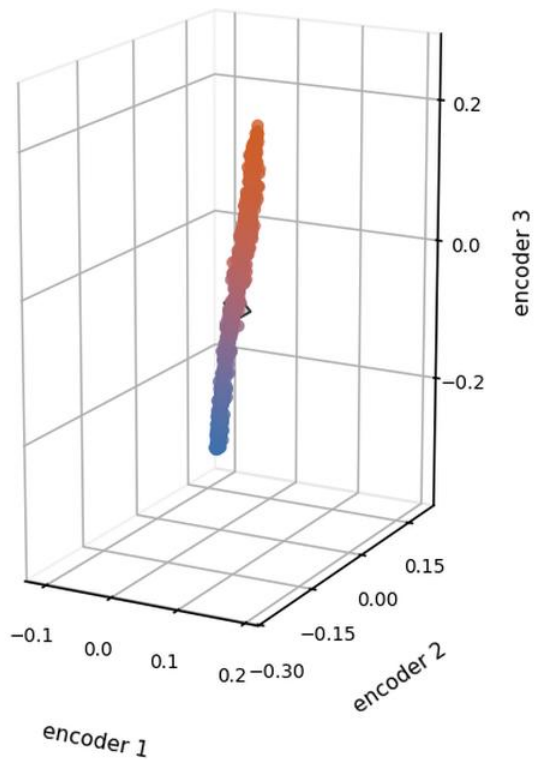


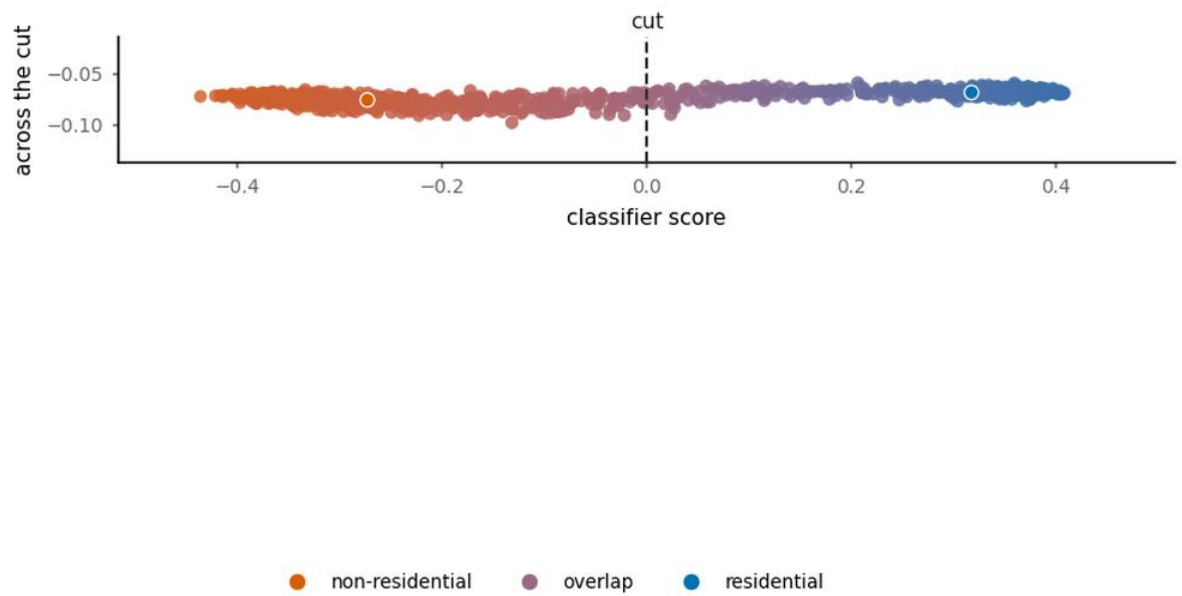
APPENDIX · CLUSTER
NYC-fold validation · Denver W

Gap 3.74

Denver: residential and non-residential are apart

Validation tiles while New York is held out. n=743 (380 non-residential, 363 residential). Gap 3.74. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



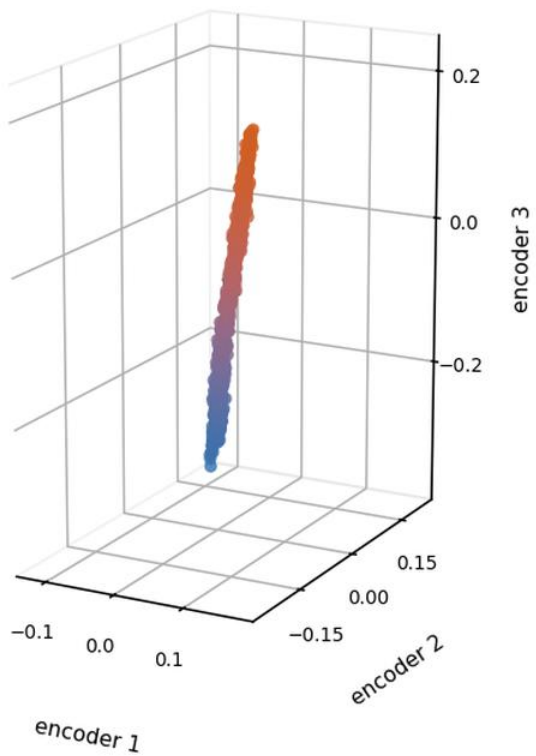


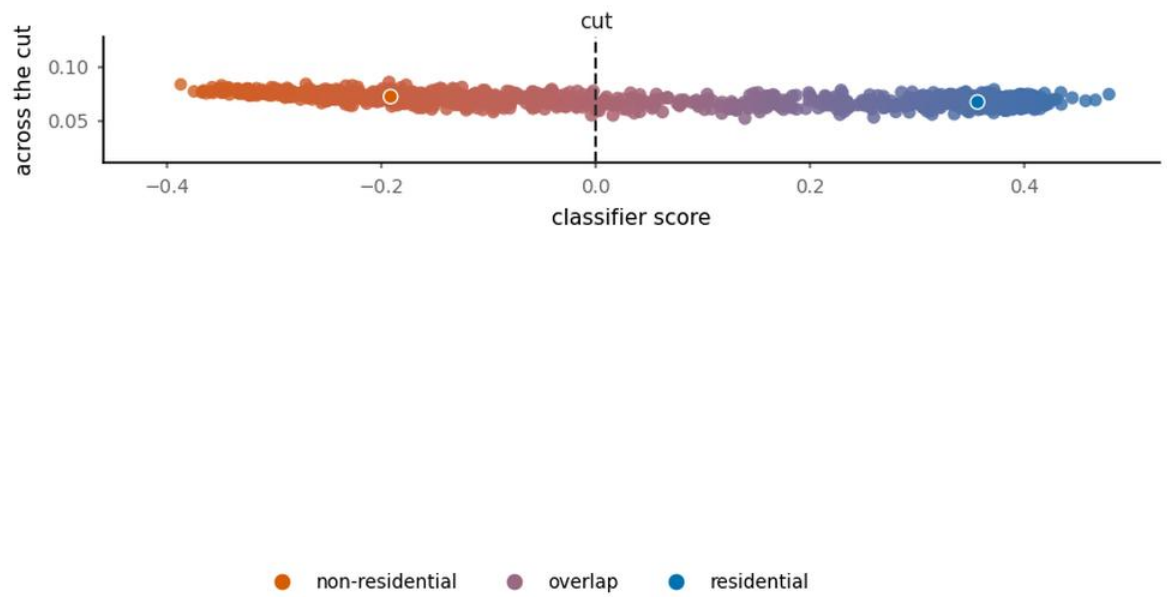
APPENDIX · CLUSTER
NYC-fold validation · LA W

Gap 3.71

Los Angeles: residential and non-residential are apart

Validation tiles while New York is held out. n=1068 (531 non-residential, 537 residential). Gap 3.71. Mauve is where both centers still reach. Influence falls with distance from a center. The plane a



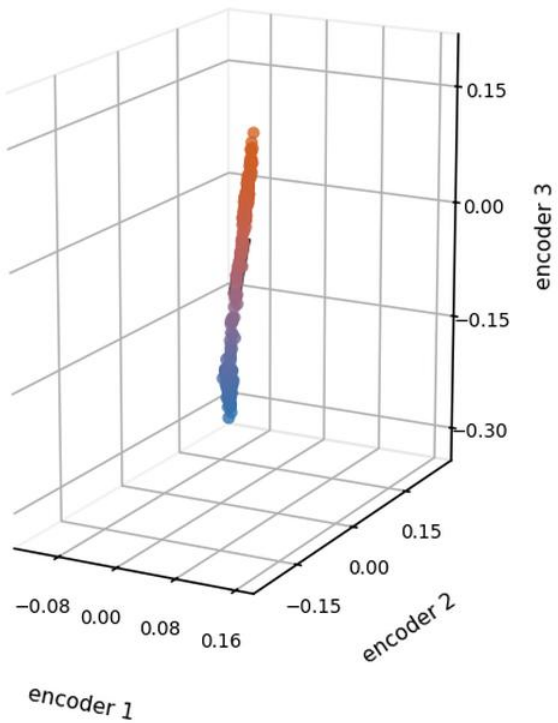


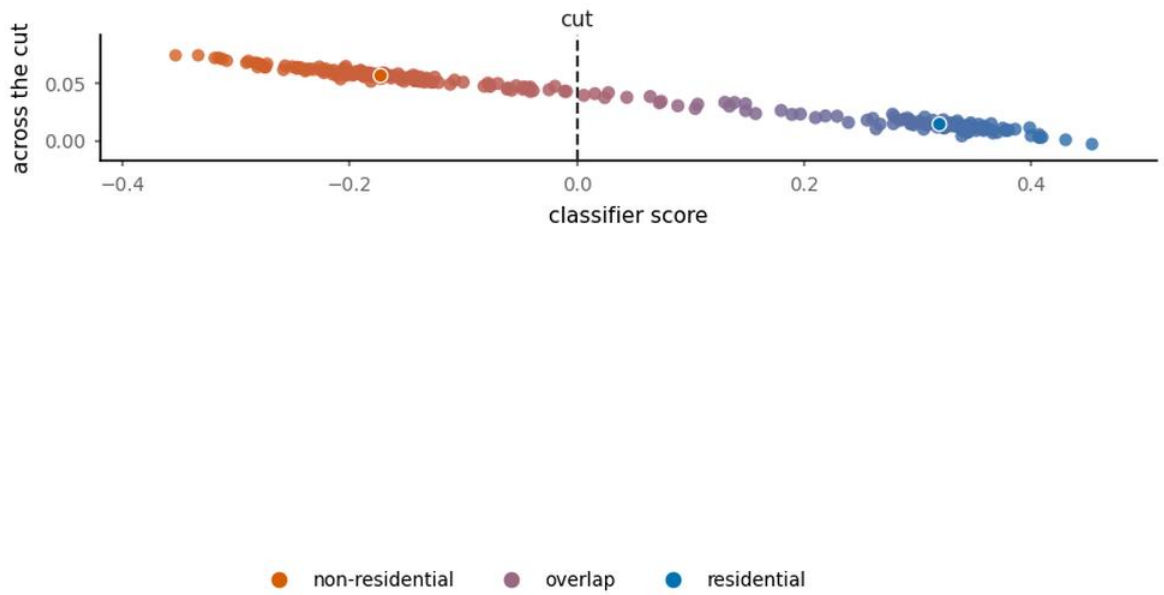
APPENDIX · CLUSTER
NYC-fold validation · Seattle NW

Gap 3.59

Seattle: residential and non-residential are apart

Validation tiles while New York is held out. n=225 (121 non-residential, 104 residential). Gap 3.59. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



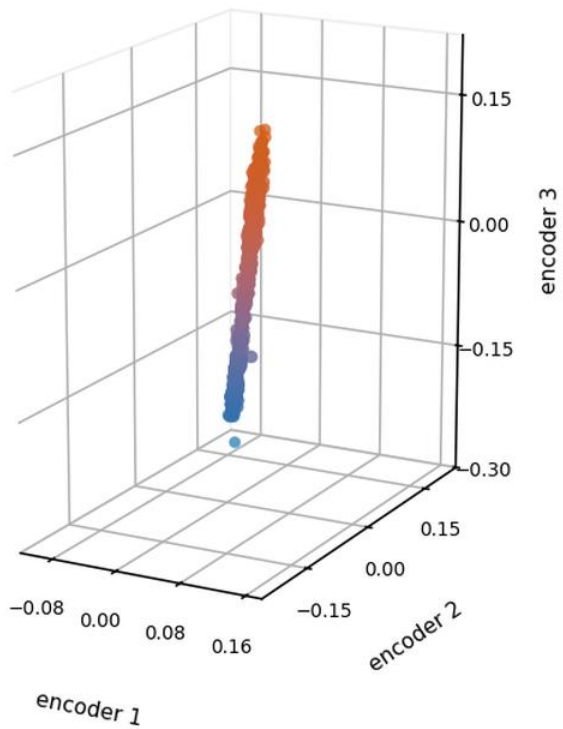


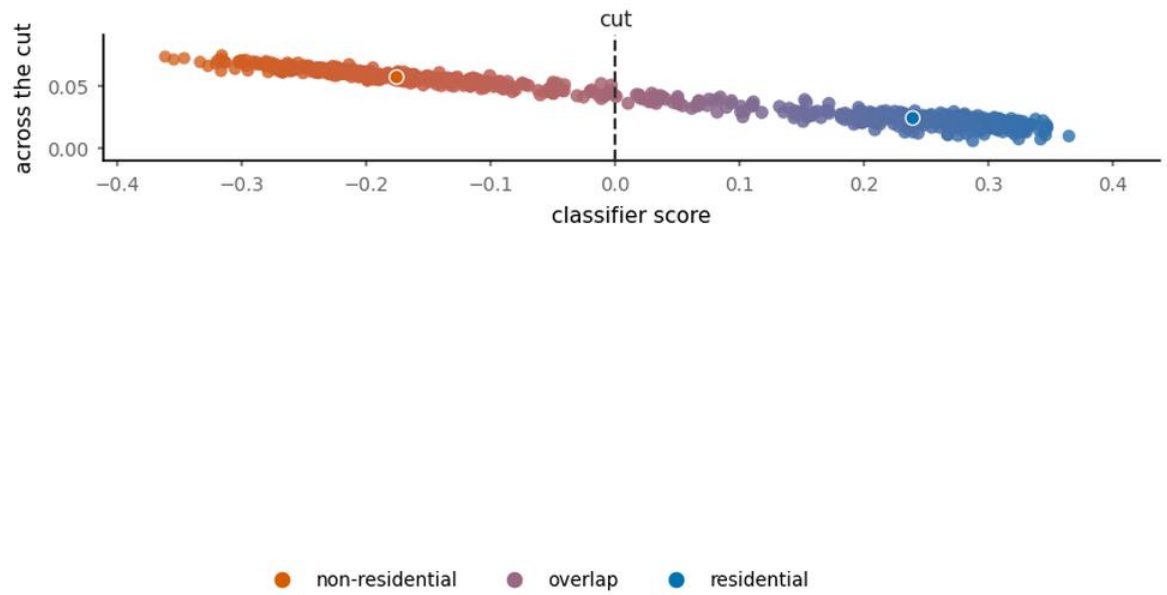
APPENDIX · CLUSTER
NYC-fold validation · Houston S

Gap 2.80

Houston: residential and non-residential are apart

Validation tiles while New York is held out. n=637 (325 non-residential, 312 residential). Gap 2.80. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



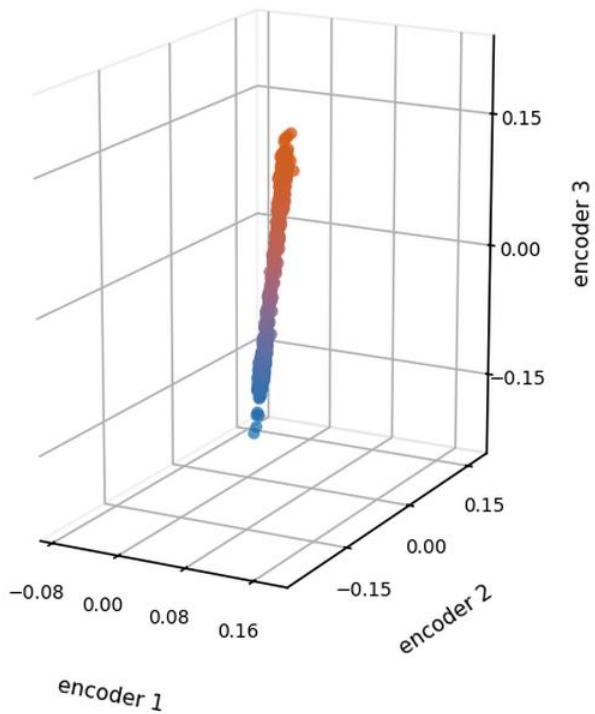


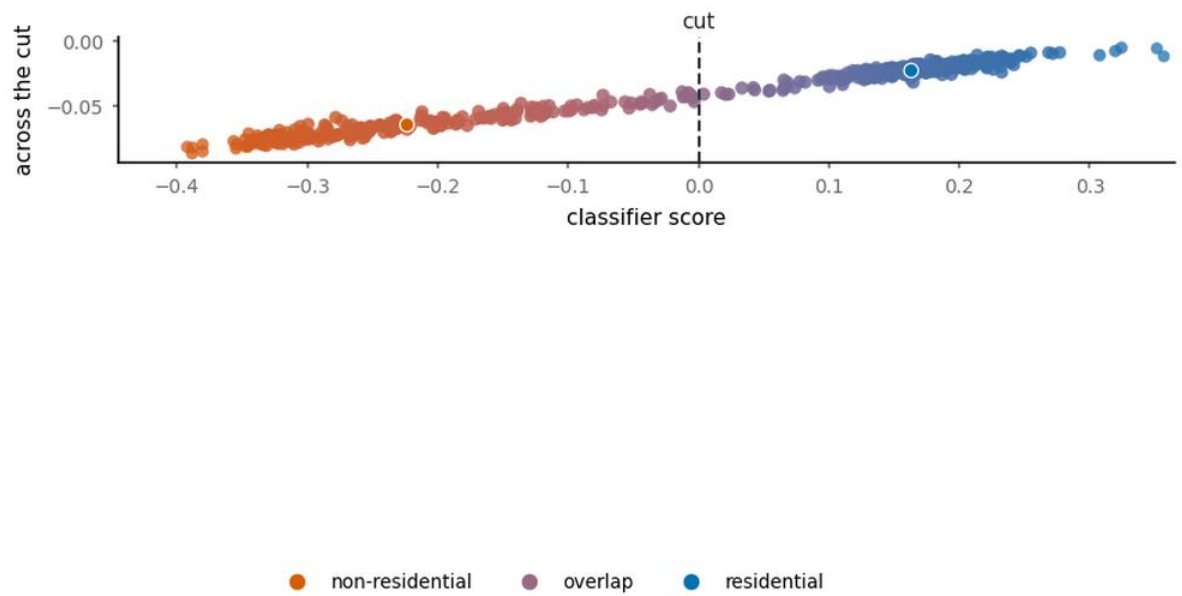
APPENDIX · CLUSTER
NYC-fold validation · Chicago MW

Gap 2.01

Chicago: the two classes touch

Validation tiles while New York is held out. n=505 (259 non-residential, 246 residential). Gap 2.01. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an



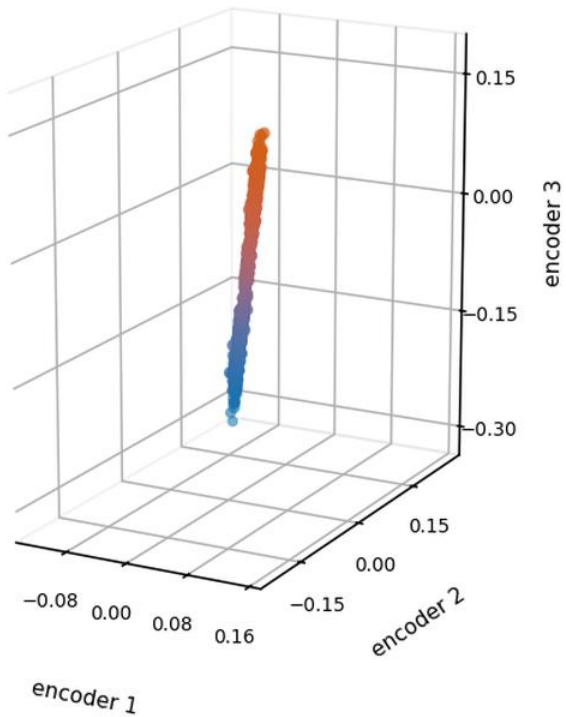


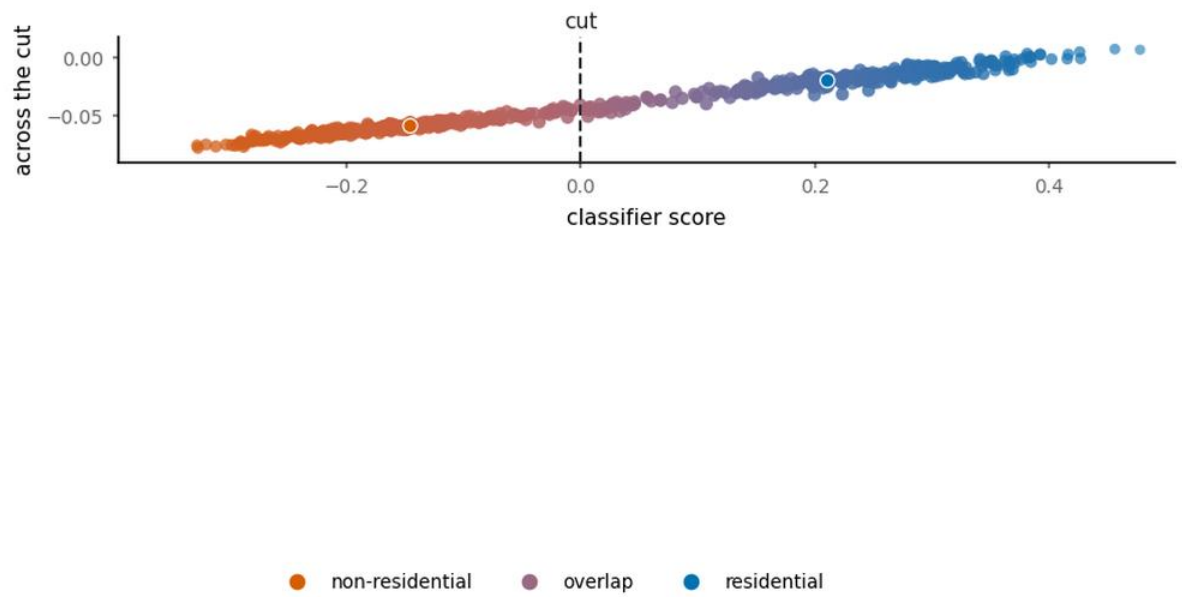
APPENDIX · CLUSTER
NYC-fold validation · Boston NE

Gap 1.90

Boston: the two classes touch

Validation tiles while New York is held out. n=550 (271 non-residential, 279 residential). Gap 1.90. Mauve is where both centers still reach. Influence falls with distance from a center. The plane an





APPENDIX · CLUSTER
NYC-fold validation · SF W

Gap 1.82

San Francisco: the two classes touch

Validation tiles while New York is held out. n=111 (50 non-residential, 61 residential). Gap 1.82. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and t

